

Single-view metrology



R. Magritte, *Personal Values*, 1952

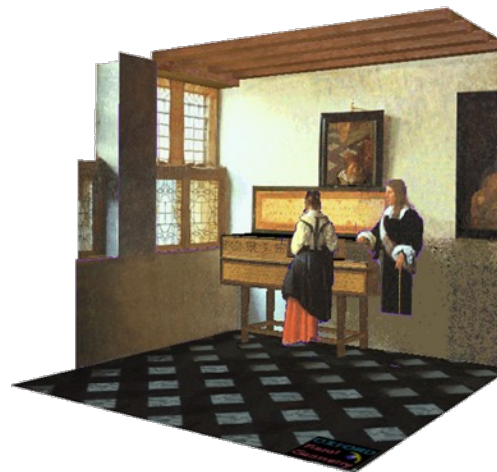
Many slides adapted from
S. Seitz, D. Hoiem

Single-view metrology

- Given a certain set of assumptions, can we recover 3D information from a single image?

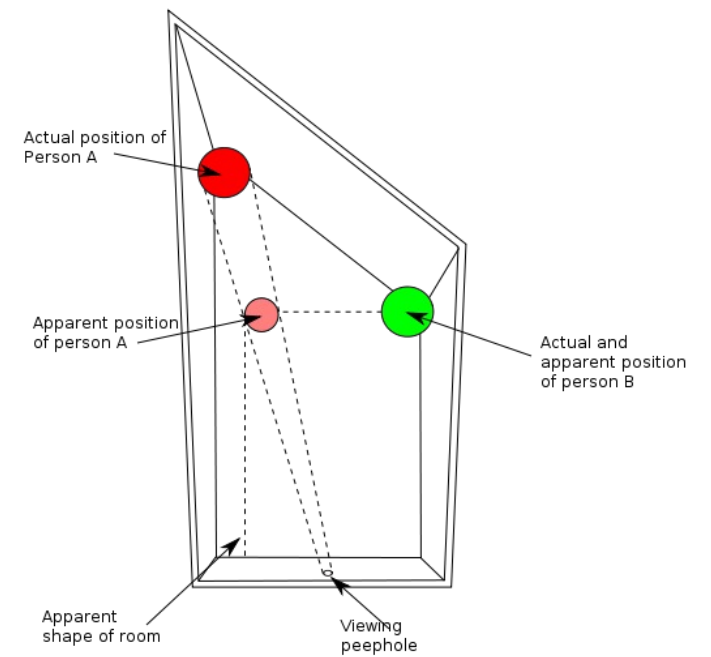


J. Vermeer, Music Lesson, 1662



A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings.](#)
Proc. Computers and the History of Art, 2002

Reminder: Beware!

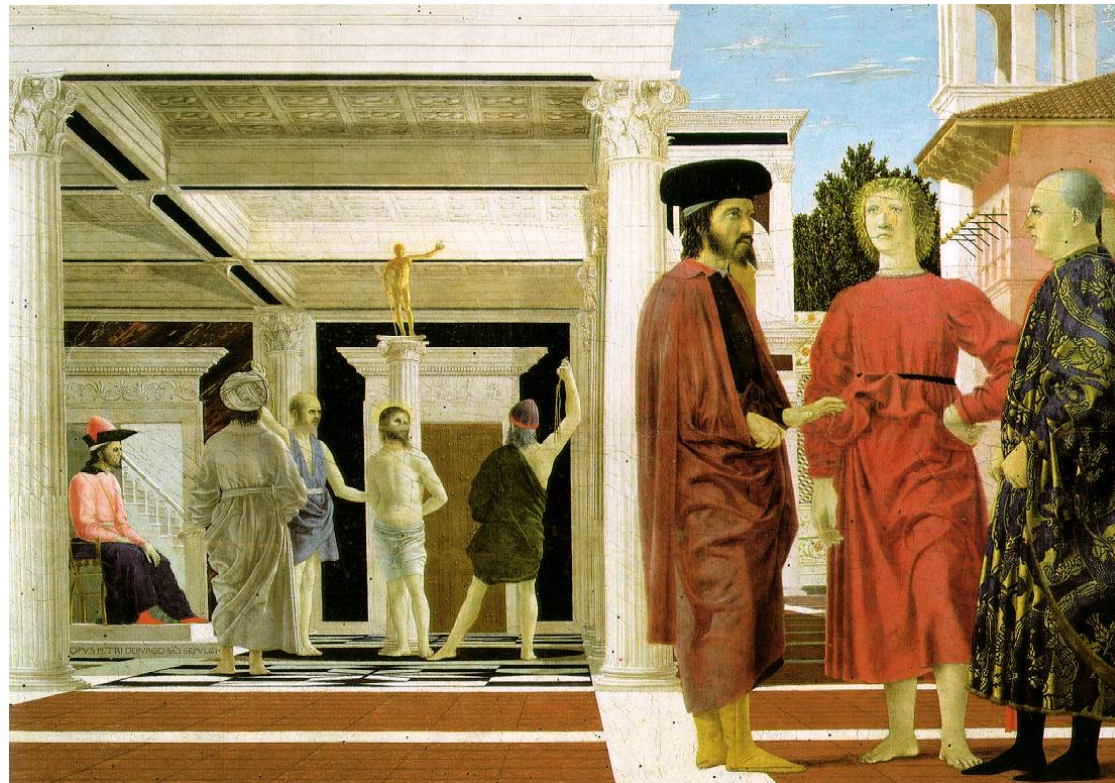


http://en.wikipedia.org/wiki/Ames_room

Outline

- Vanishing points and vanishing lines
- Camera calibration using vanishing points
- Measurements from a single image
- Applications of single-view metrology

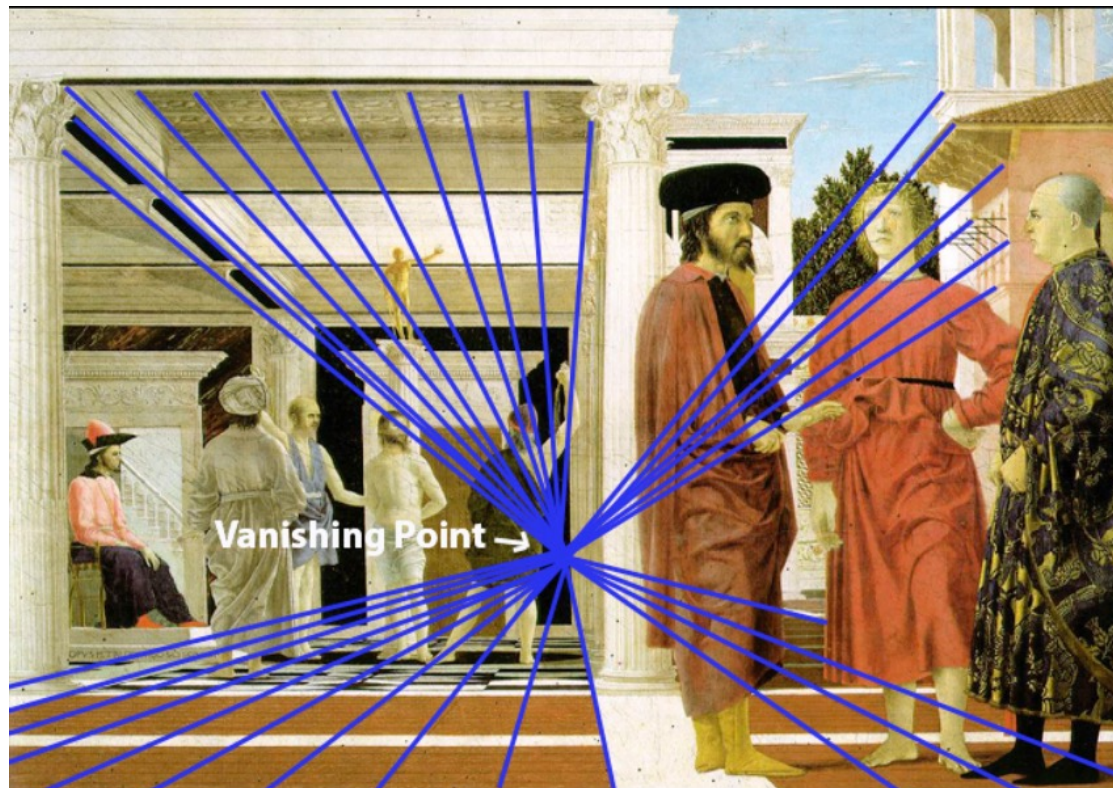
Vanishing points



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Vanishing points

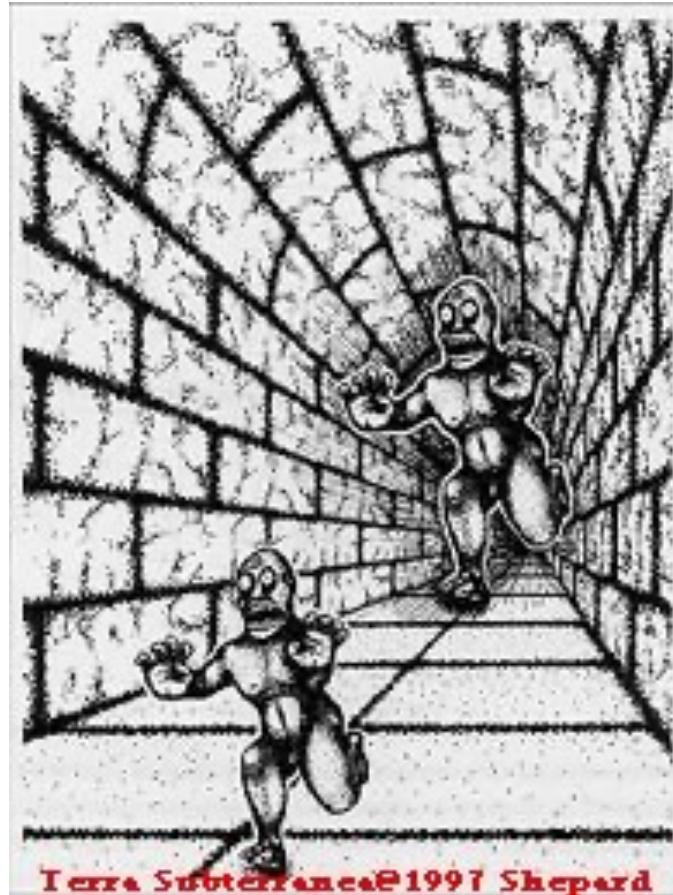
- Parallel lines meet at a *vanishing point*



Piero della Francesca, *Flagellation of Christ*, 1455-1460

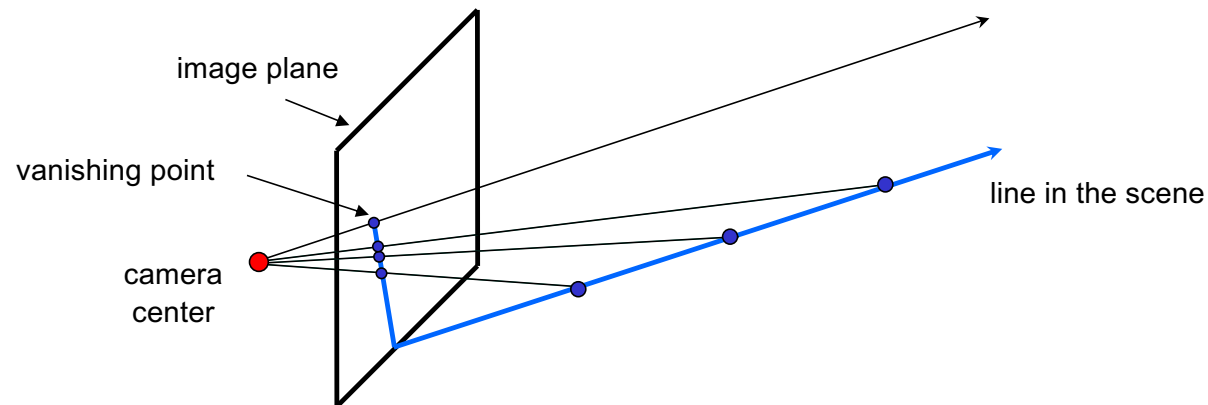
[Figure source](#)

Converging lines are a powerful perspective cue



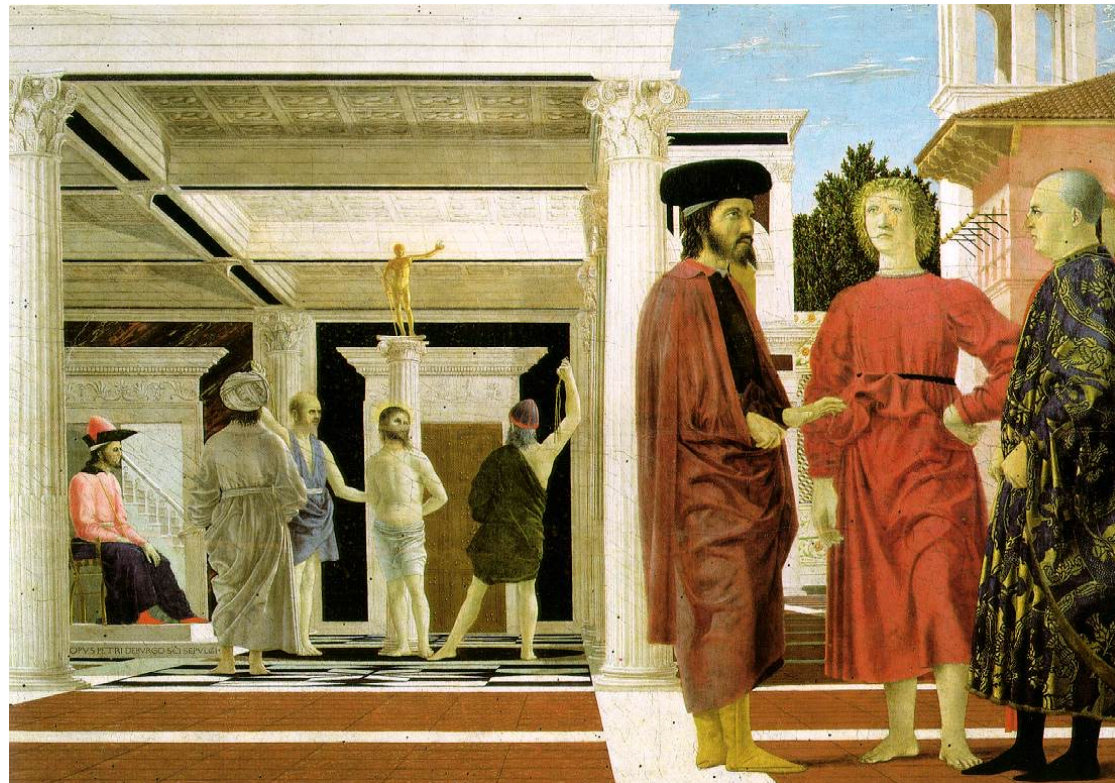
Slide by Steve Seitz

Constructing the vanishing point of a line



- What about another line going in the same direction?

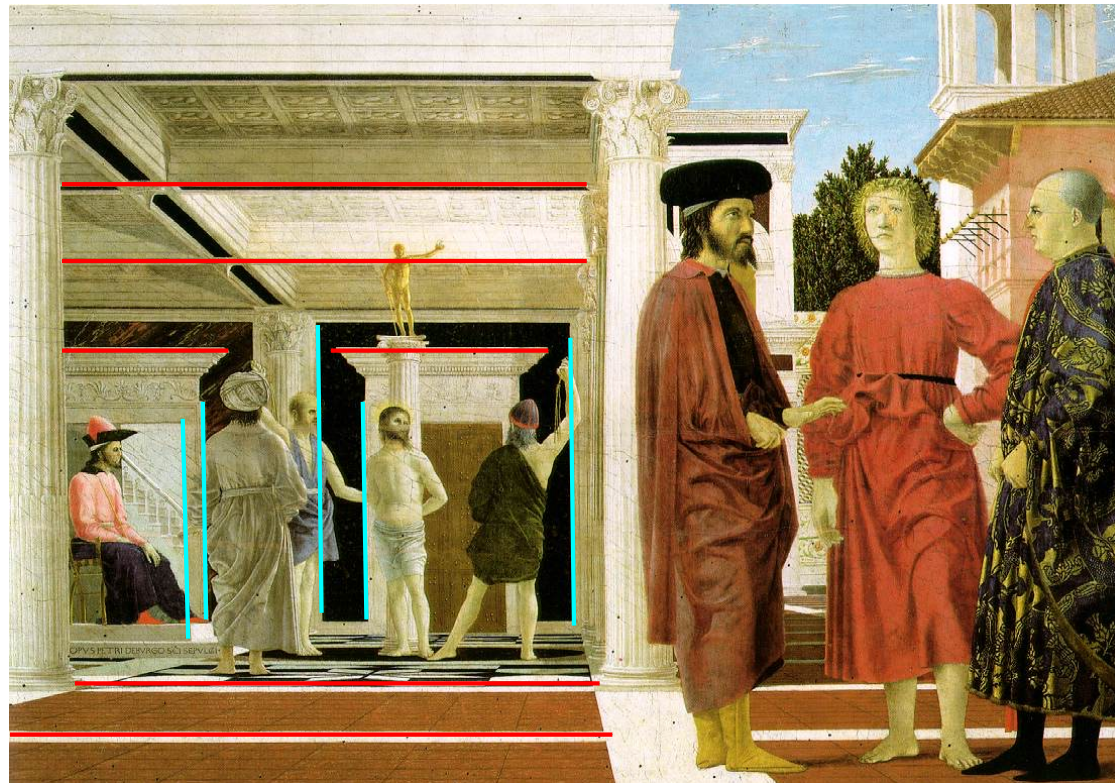
Do *all* parallel lines converge to a vanishing point?



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Do *all* parallel lines converge to a vanishing point?

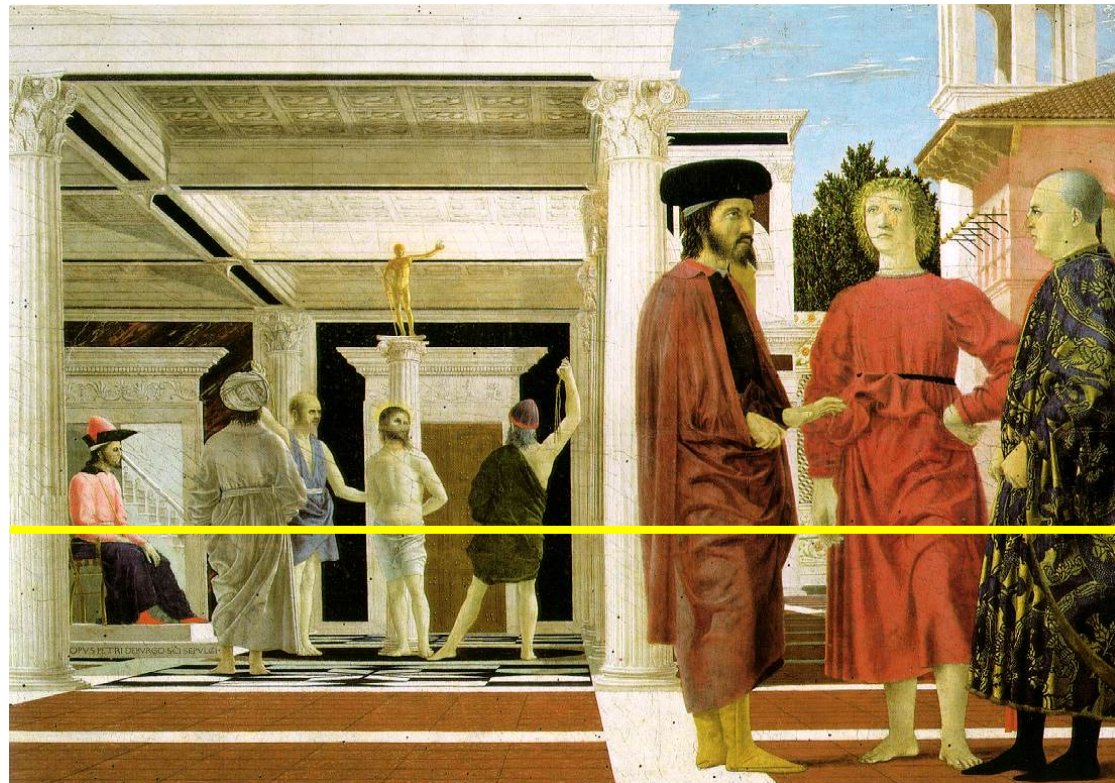
- Not parallel lines that are also parallel to the image plane!



Piero della Francesca, *Flagellation of Christ*, 1455-1460

Vanishing lines of planes

- Each family of parallel planes is associated with a *vanishing line* in the image



Vanishing line of
the ground plane

Piero della Francesca, *Flagellation of Christ*, 1455-1460

Constructing the vanishing line of a plane

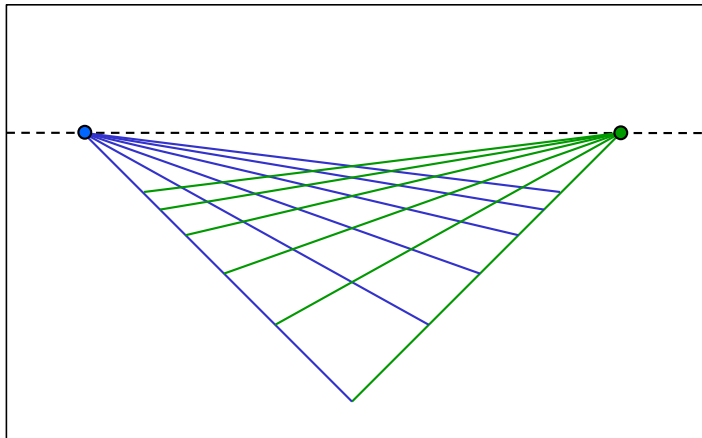


Figure source: S. Seitz

Constructing the vanishing line of a plane

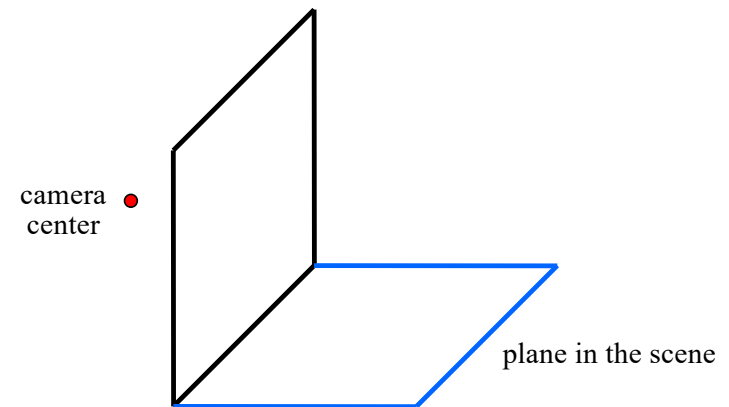
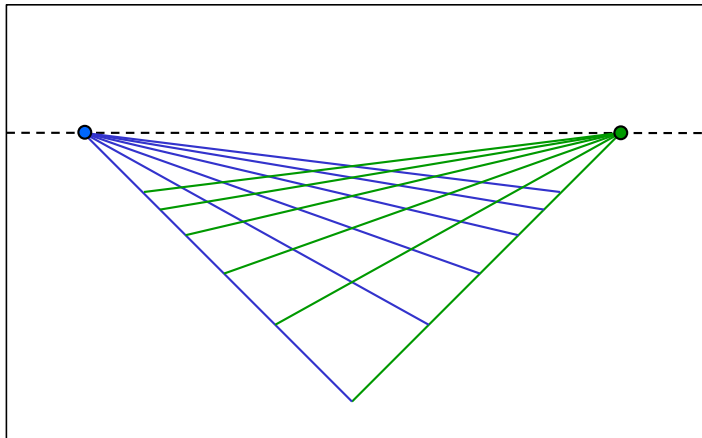
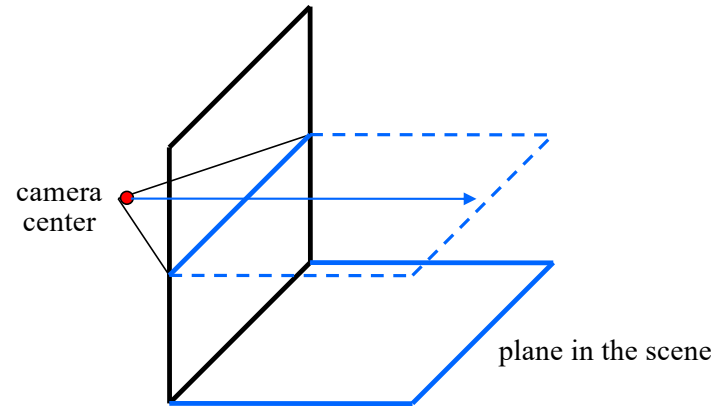


Figure source: S. Seitz



Horizon: Vanishing line of the ground plane

- What can the horizon tell us about the relative height of scene points and the camera?

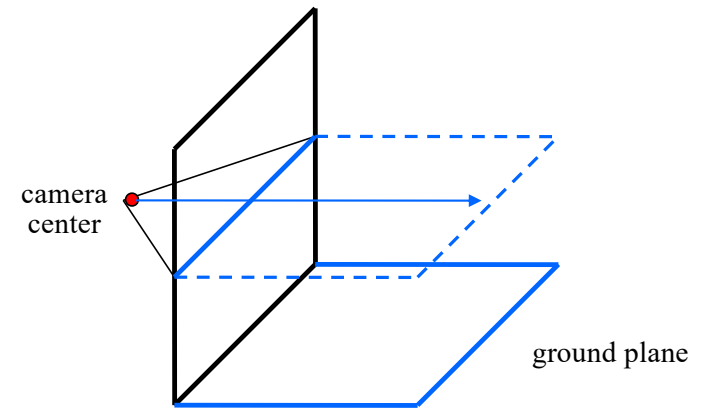
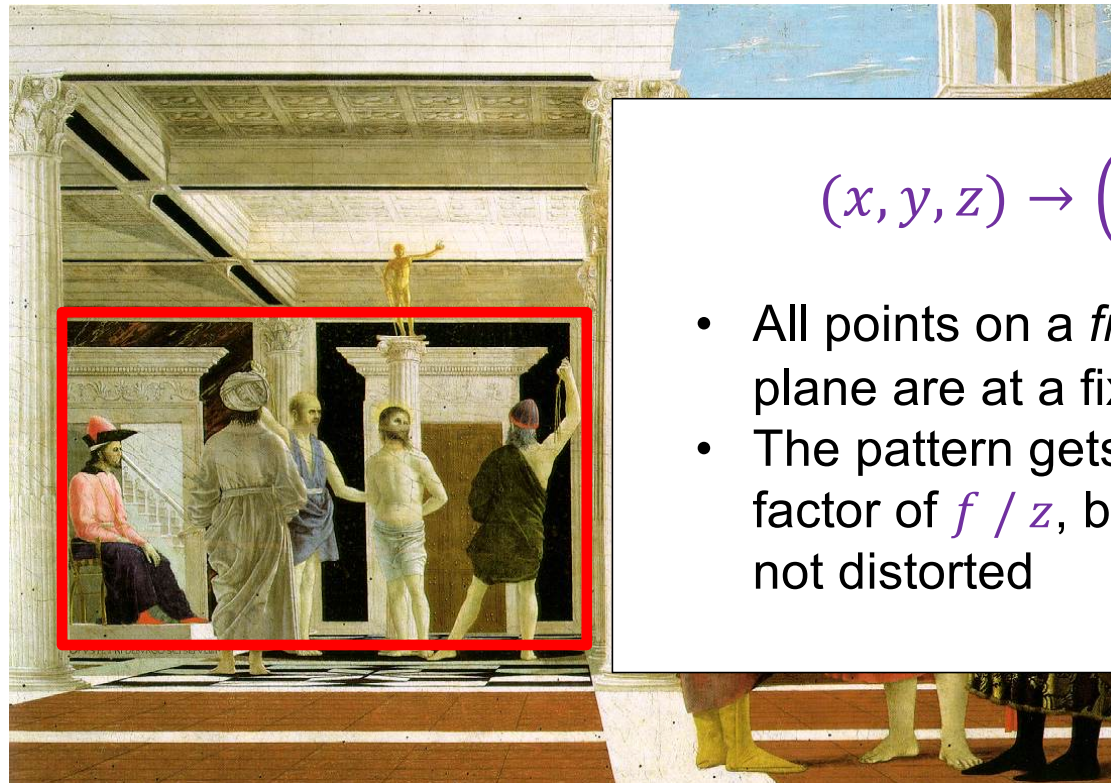


Image source: S. Seitz

Projection of planes

- Are there planes that don't have vanishing lines?



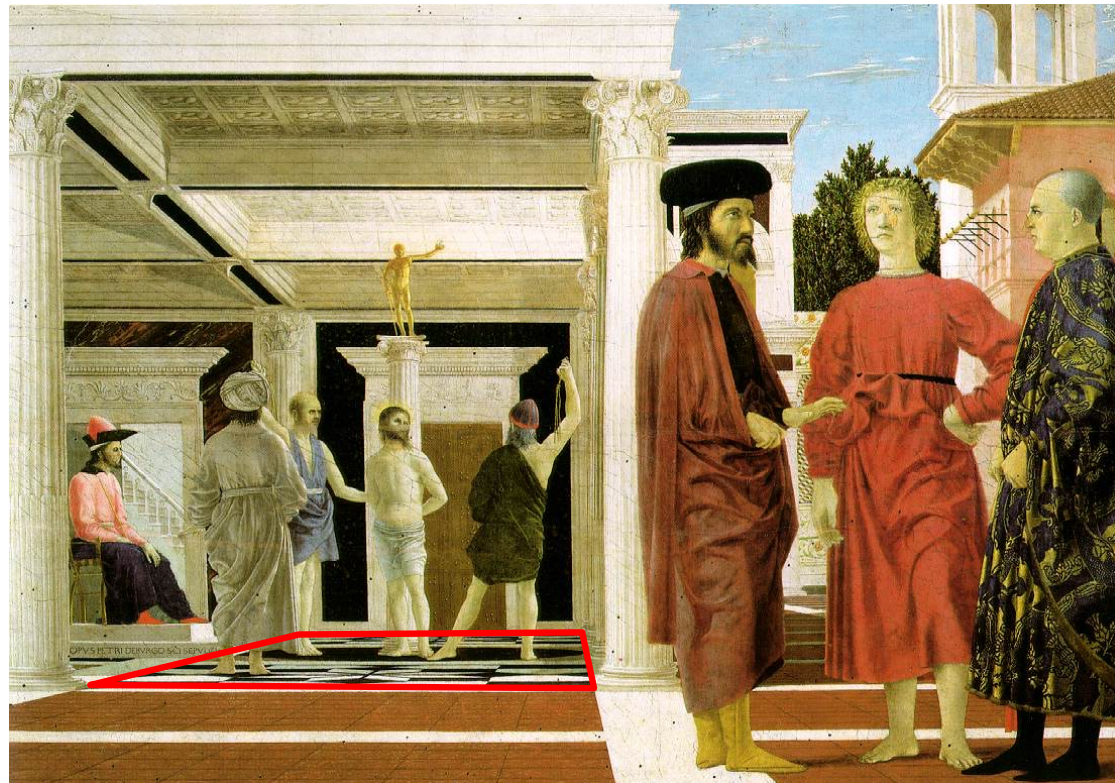
$$(x, y, z) \rightarrow \left(f \frac{x}{z}, f \frac{y}{z}\right)$$

- All points on a *fronto-parallel* plane are at a fixed depth z
- The pattern gets scaled by a factor of f / z , but is otherwise not distorted

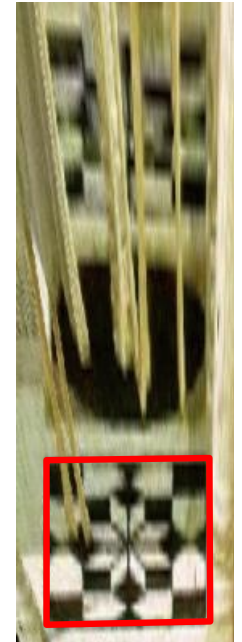
Piero della Francesca, *Flagellation of Christ*, 1455-1460

Projection of planes

- Patterns on *non-fronto-parallel* planes are distorted by a 2D homography



Piero della Francesca, *Flagellation of Christ*, 1455-1460



Outline

- Vanishing points and vanishing lines
- Camera calibration using vanishing points

Recall: Camera calibration

- Given n points with known 3D coordinates X_i and known image projections x_i , estimate the camera matrix $P = K[R | t]$



Image credit: J. Hays

Known 2D image coords	Known 3D locations
--------------------------	-----------------------

880 214	312.747 309.140 30.086
43 203	305.796 311.649 30.356
270 197	307.694 312.358 30.418
886 347	310.149 307.186 29.298
745 302	311.937 310.105 29.216
943 128	311.202 307.572 30.682
476 590	307.106 306.876 28.660
419 214	309.317 312.490 30.230
317 335	307.435 310.151 29.318
783 521	308.253 306.300 28.881
235 427	306.650 309.301 28.905
665 429	308.069 306.831 29.189
655 362	309.671 308.834 29.029
427 333	308.255 309.955 29.267
412 415	307.546 308.613 28.963
746 351	311.036 309.206 28.913
434 415	307.518 308.175 29.069
525 234	309.950 311.262 29.990
716 308	312.160 310.772 29.080
602 187	311.988 312.709 30.514

Camera calibration using vanishing points

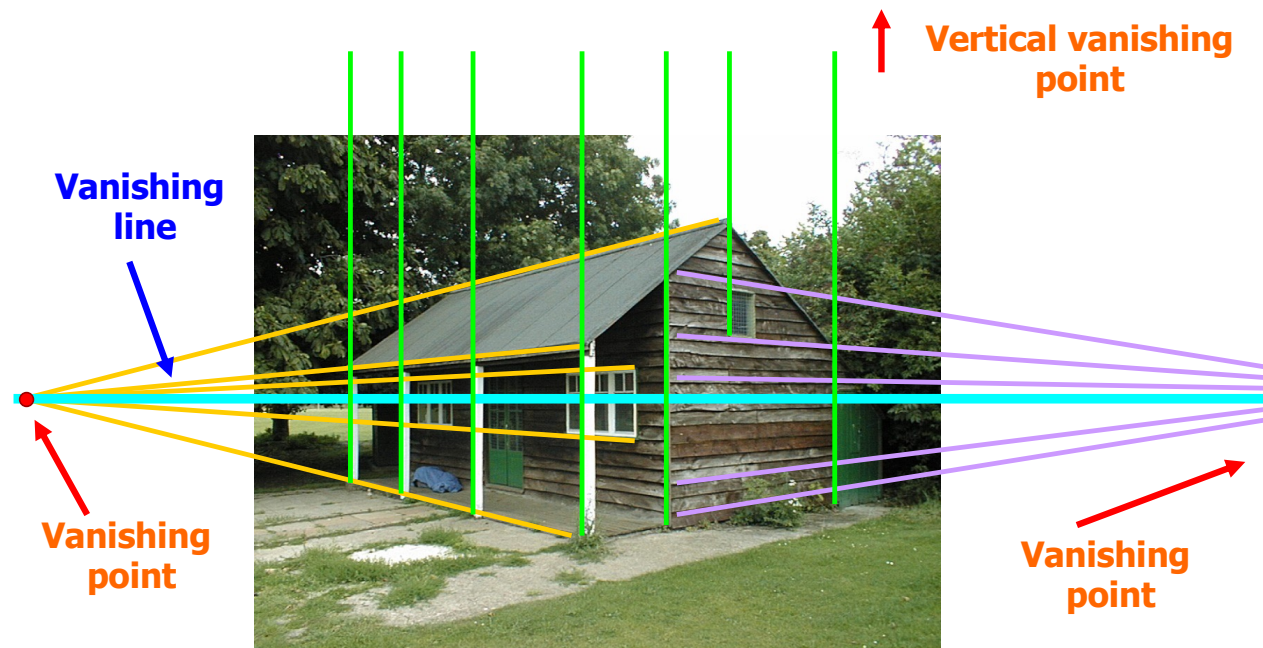
- If world coordinates of reference 3D points are not known, we can use vanishing points of three orthogonal directions



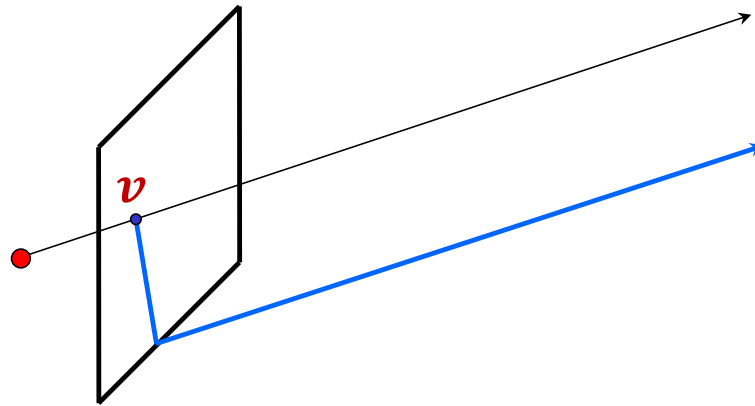
Source: A. Efros, A. Criminisi

Camera calibration using vanishing points

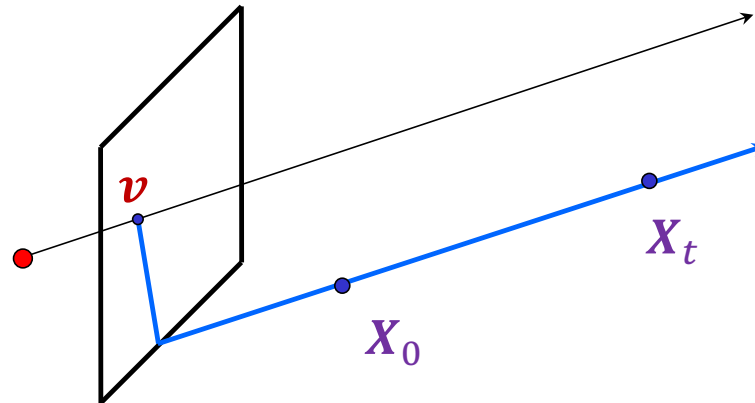
- If world coordinates of reference 3D points are not known, we can use vanishing points of three orthogonal directions



Computing vanishing points



Computing vanishing points

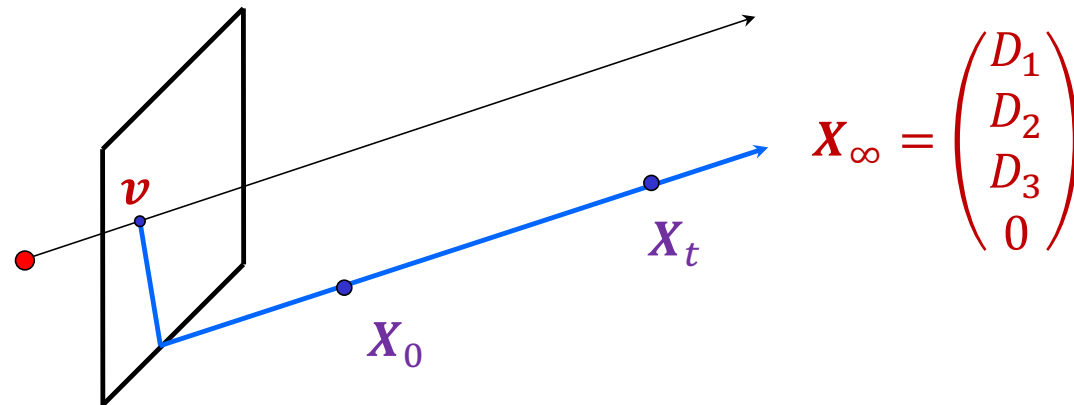


- Let's parameterize the line using point $X_0 = (X_0, Y_0, Z_0, 1)^T$ and direction vector $D = (D_1, D_2, D_3)^T$:

$$X_t = \begin{pmatrix} X_0 + tD_1 \\ Y_0 + tD_2 \\ Z_0 + tD_3 \\ 1 \end{pmatrix} \cong \begin{pmatrix} X_0/t + D_1 \\ Y_0/t + D_2 \\ Z_0/t + D_3 \\ 1/t \end{pmatrix} \quad X_\infty = \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ 0 \end{pmatrix}$$

- X_∞ is a *point at infinity*, v is its projection: $v \cong PX_\infty$

Computing vanishing points

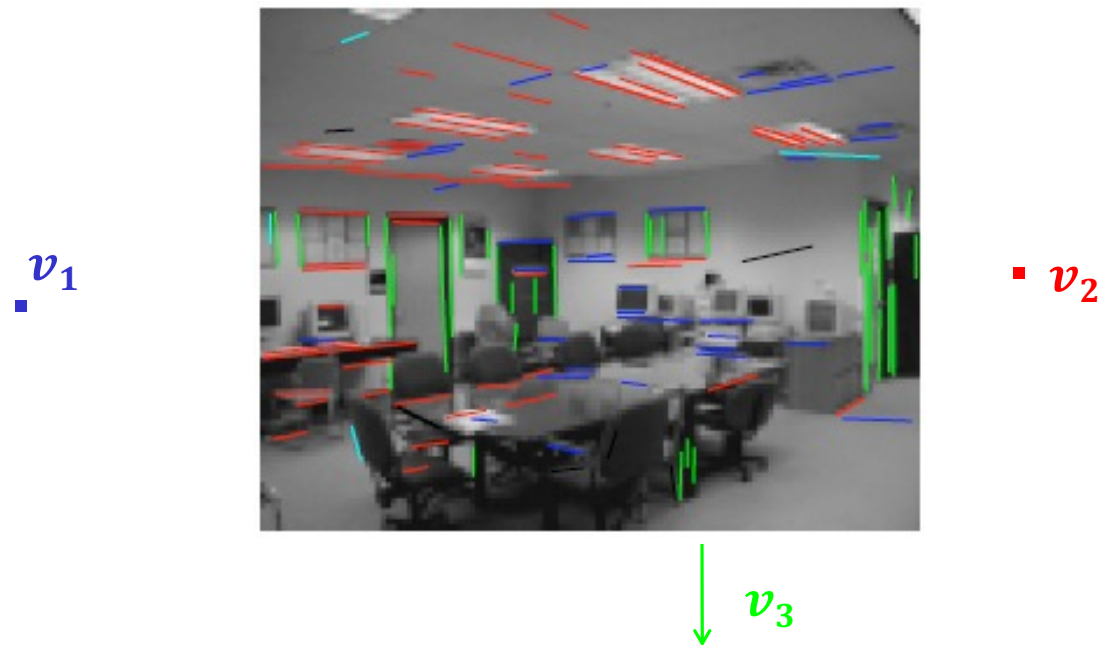


- X_∞ is a *point at infinity*, v is its projection: $v \cong PX_\infty$
- In the image, vanishing points can be *finite* or *infinite*
- In canonical camera coordinates:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ 0 \end{bmatrix} = \begin{bmatrix} D_1 \\ D_2 \\ D_3 \end{bmatrix} \quad D_3 = 0: \text{fronto-parallel direction}$$

Calibration from vanishing points

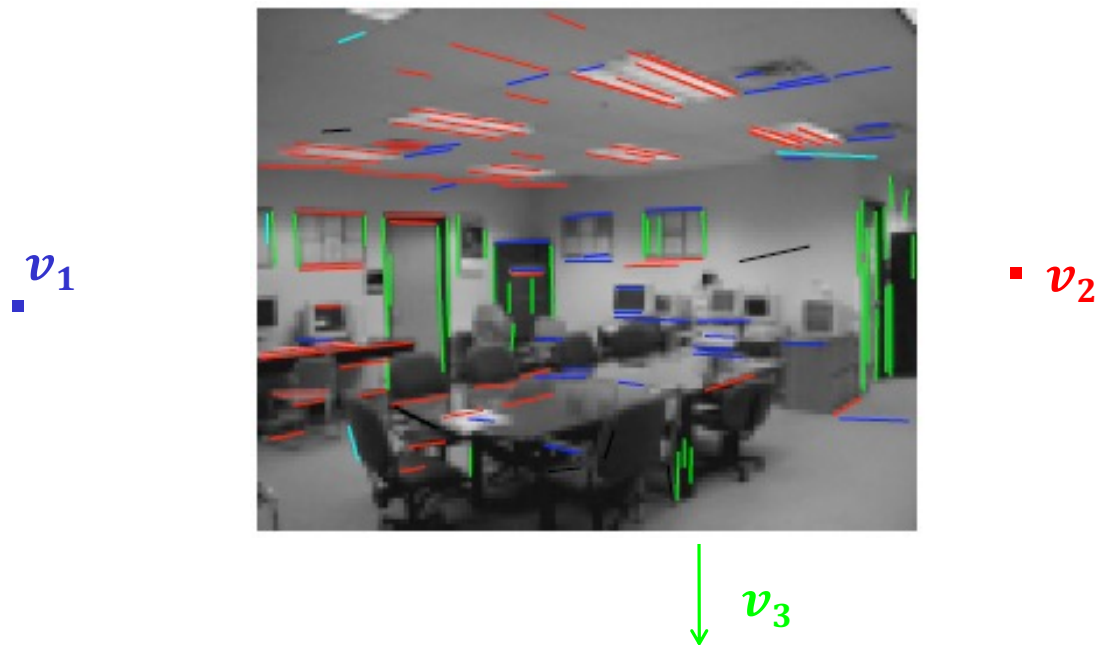
- Consider a scene with three orthogonal vanishing directions:



- In this case, v_1 , v_2 are finite vanishing points and v_3 is an infinite vanishing point

Calibration from vanishing points

- Consider a scene with three orthogonal vanishing directions:



- We can align the world coordinate system with these directions

Projection of the world coordinate system

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}$$

$$\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4$$

Projection of the world coordinate system

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{p}_1$$

$\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4$

Projection of the world coordinate system

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \mathbf{p}_2$$

$\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4$

Projection of the world coordinate system

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \mathbf{p}_3$$

$$\underbrace{\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3}_{\text{Vanishing points in } x, y, z \text{ directions}}$$

(i.e., $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$)

Projection of the world coordinate system

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{p}_4$$

$\underbrace{\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3}_{\text{Vanishing points in } x, y, z \text{ directions (i.e., } \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)}$

$\mathbf{p}_4$

Projection of world origin

- Problem: this only gives us the four columns up to independent scale factors, additional constraints needed to solve for them

Calibration from vanishing points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \mathbf{v}_i \cong \mathbf{K}[\mathbf{R}|\mathbf{t}] \begin{pmatrix} \mathbf{e}_i \\ 0 \end{pmatrix}$$

Calibration from vanishing points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \begin{aligned} \mathbf{v}_i &\cong \mathbf{K} \mathbf{R} \mathbf{e}_i \\ \mathbf{e}_i &\cong \mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_i \end{aligned}$$

- Orthogonality constraint: $\mathbf{e}_i^T \mathbf{e}_j = 0$

$$\underbrace{\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{R} \mathbf{R}^T \mathbf{K}^{-1}}_{\mathbf{e}_i^T} \underbrace{\mathbf{v}_j}_{\mathbf{e}_j} = 0$$

Calibration from vanishing points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \begin{aligned} \mathbf{v}_i &\cong \mathbf{K} \mathbf{R} \mathbf{e}_i \\ \mathbf{e}_i &\cong \mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_i \end{aligned}$$

- Orthogonality constraint: $\mathbf{e}_i^T \mathbf{e}_j = 0$

$$\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{v}_j = 0$$

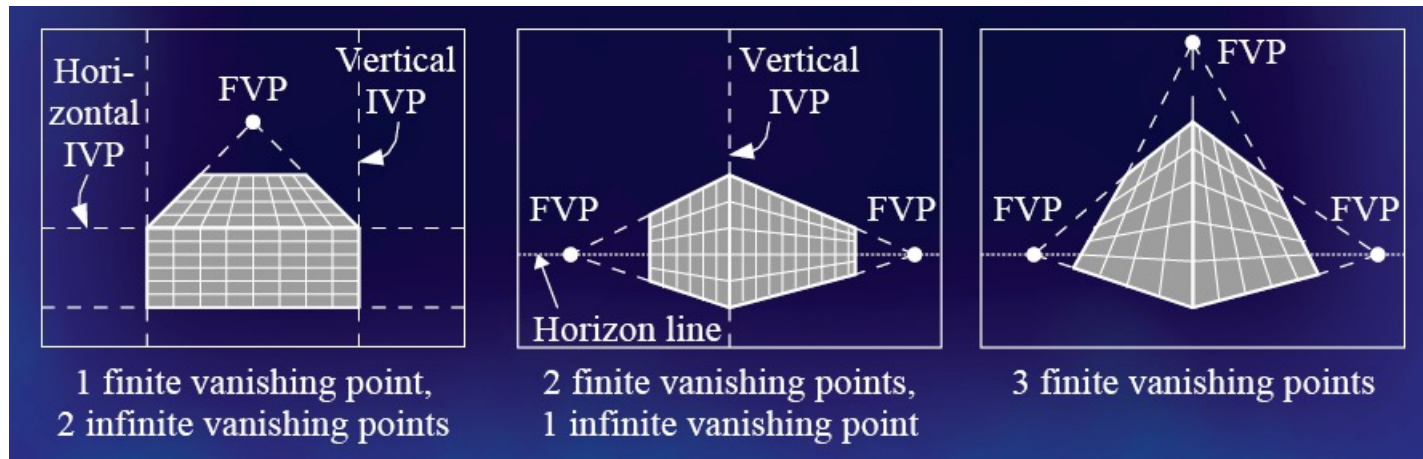
- Extrinsic parameter matrix ($\mathbf{R}$) disappears and we are left with constraints on the calibration matrix!

Calibration from vanishing points

$$\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{v}_j = 0$$

- How many constraints do we get?
 - Three: one for each pair of vanishing points
- How many unknown parameters does $\mathbf{K}$ have?
 - Three: f, p_x, p_y
- A couple of complications:
 - The constraints are nonlinear, but it's not hard to do the algebra (omitted)
 - At least two *finite* vanishing points are needed to solve for both focal length and principal point

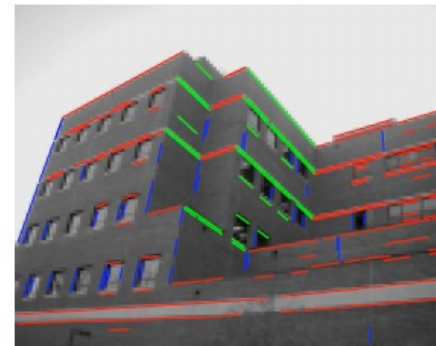
Calibration from vanishing points



Cannot recover focal length, principal point is the third vanishing point



Can solve for focal length, principal point



Rotation from vanishing points

- Constraints on vanishing points: $\mathbf{v}_i \cong \mathbf{K}\mathbf{R}\mathbf{e}_i$
- We just used orthogonality constraints to solve for $\mathbf{K}$
- Now we have:

$$\mathbf{K}^{-1}\mathbf{v}_i \cong \mathbf{R}\mathbf{e}_i$$

$$\text{Notice: } \mathbf{R}\mathbf{e}_1 = [\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{r}_3] \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \mathbf{r}_1$$

$$\text{Thus, } \mathbf{r}_i = \mathbf{R}\mathbf{e}_i \cong \mathbf{K}^{-1}\mathbf{v}_i$$

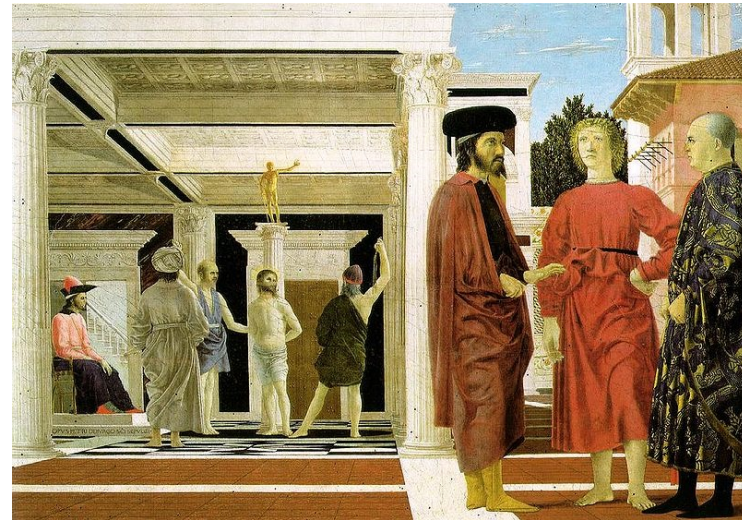
- The scale ambiguity goes away since we require $\|\mathbf{r}_i\|^2 = 1$

Calibration from vanishing points: Summary

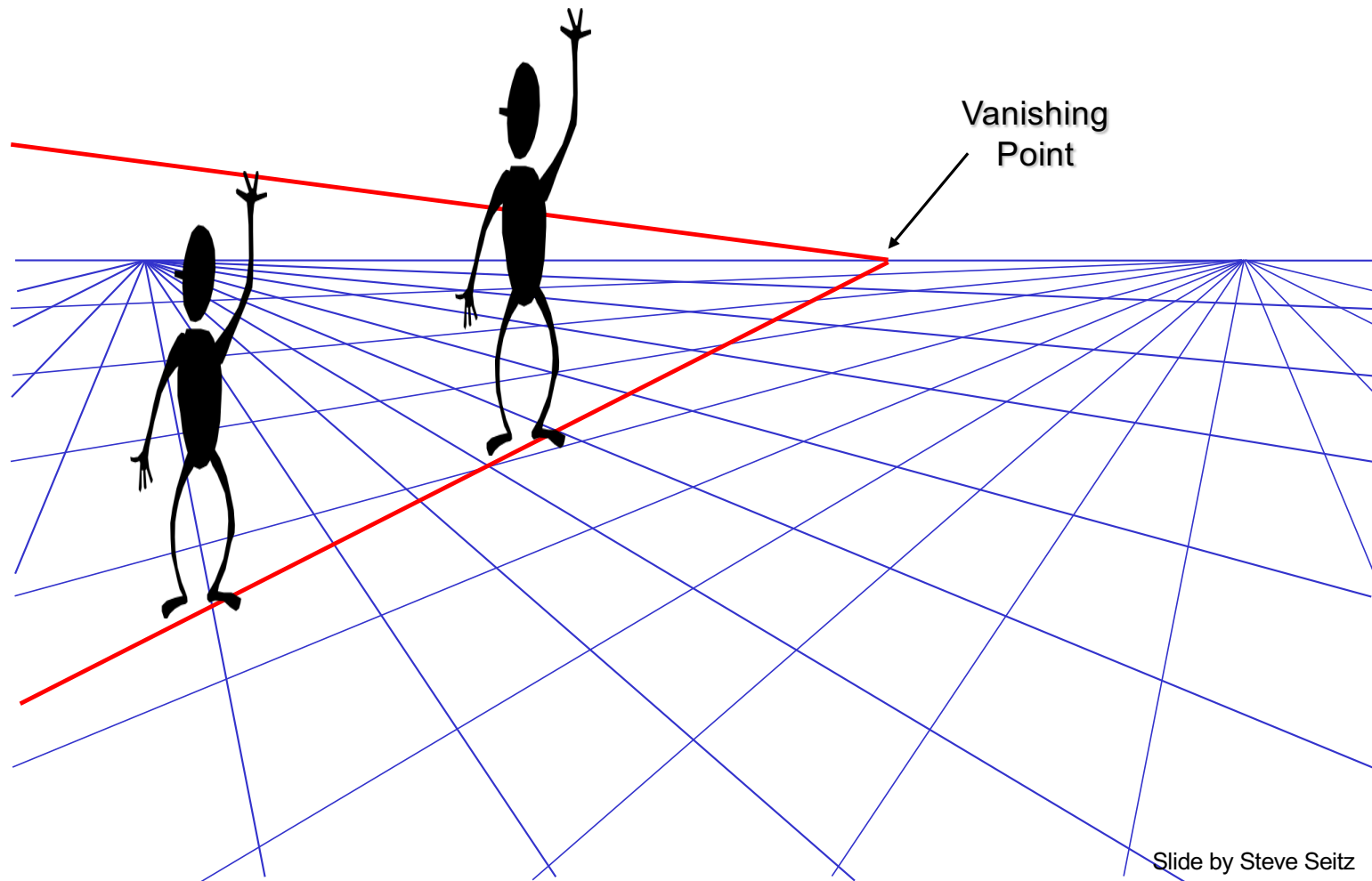
1. Solve for intrinsic parameters (focal length, principal point) using three orthogonal vanishing points
 2. Get extrinsic parameters (rotation) directly from vanishing points once calibration matrix is known
- Advantages
 - No need for calibration chart, 2D-3D correspondences
 - Could be completely automatic
 - Disadvantages
 - Only applies to certain kinds of scenes
 - It is tricky to accurately localize vanishing points
 - Need at least two finite vanishing points

Outline

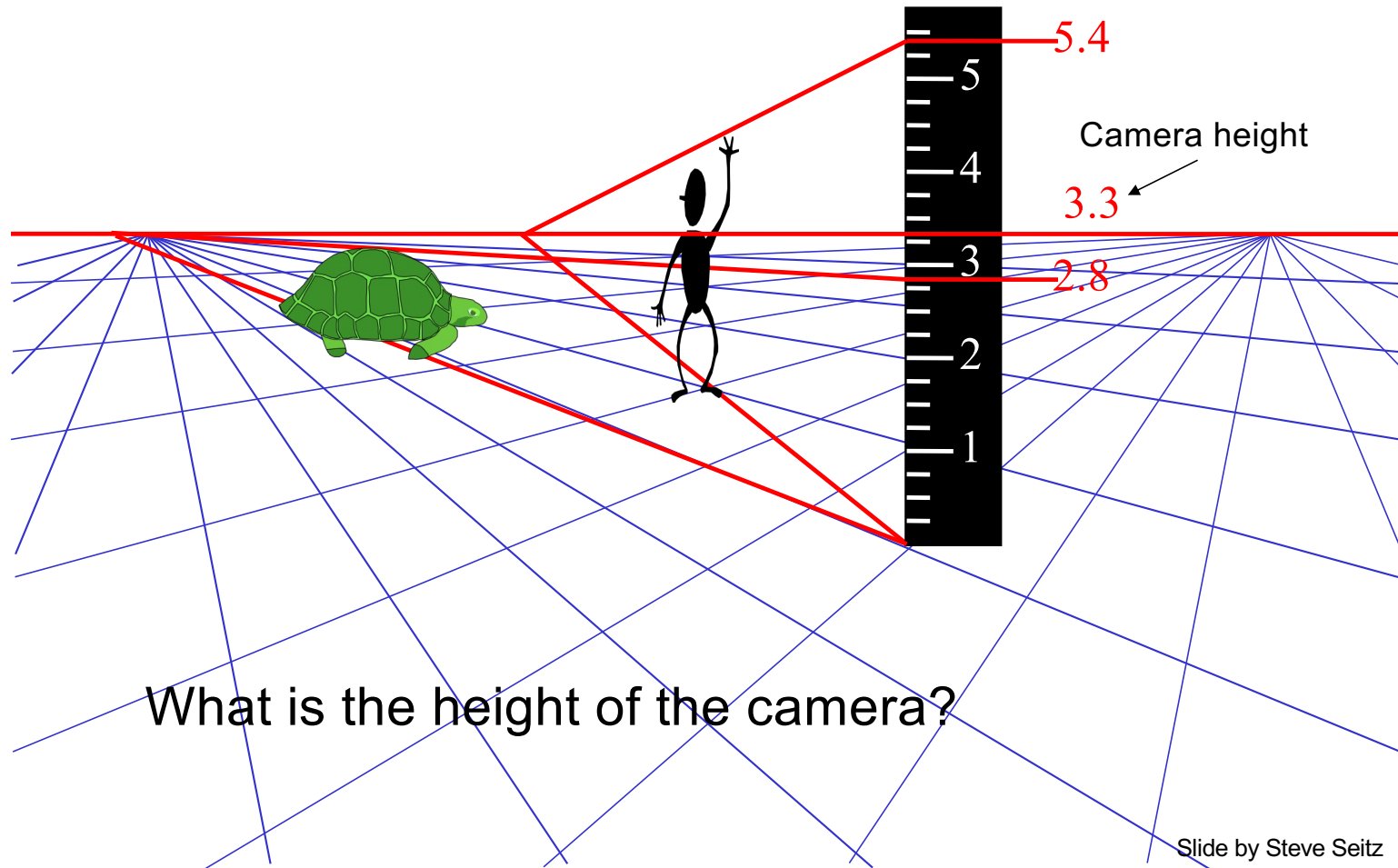
- Vanishing points and lines
- Camera calibration using vanishing points
- Measurements from a single image
 - Measuring height above the ground plane
 - Measuring within planes



Comparing heights



Measuring height with a ruler



Recall: Horizon

- What can the horizon tell us about the relative height of scene points and the camera?

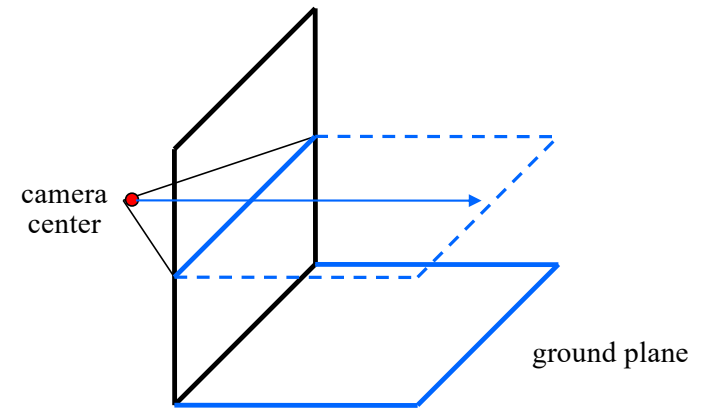
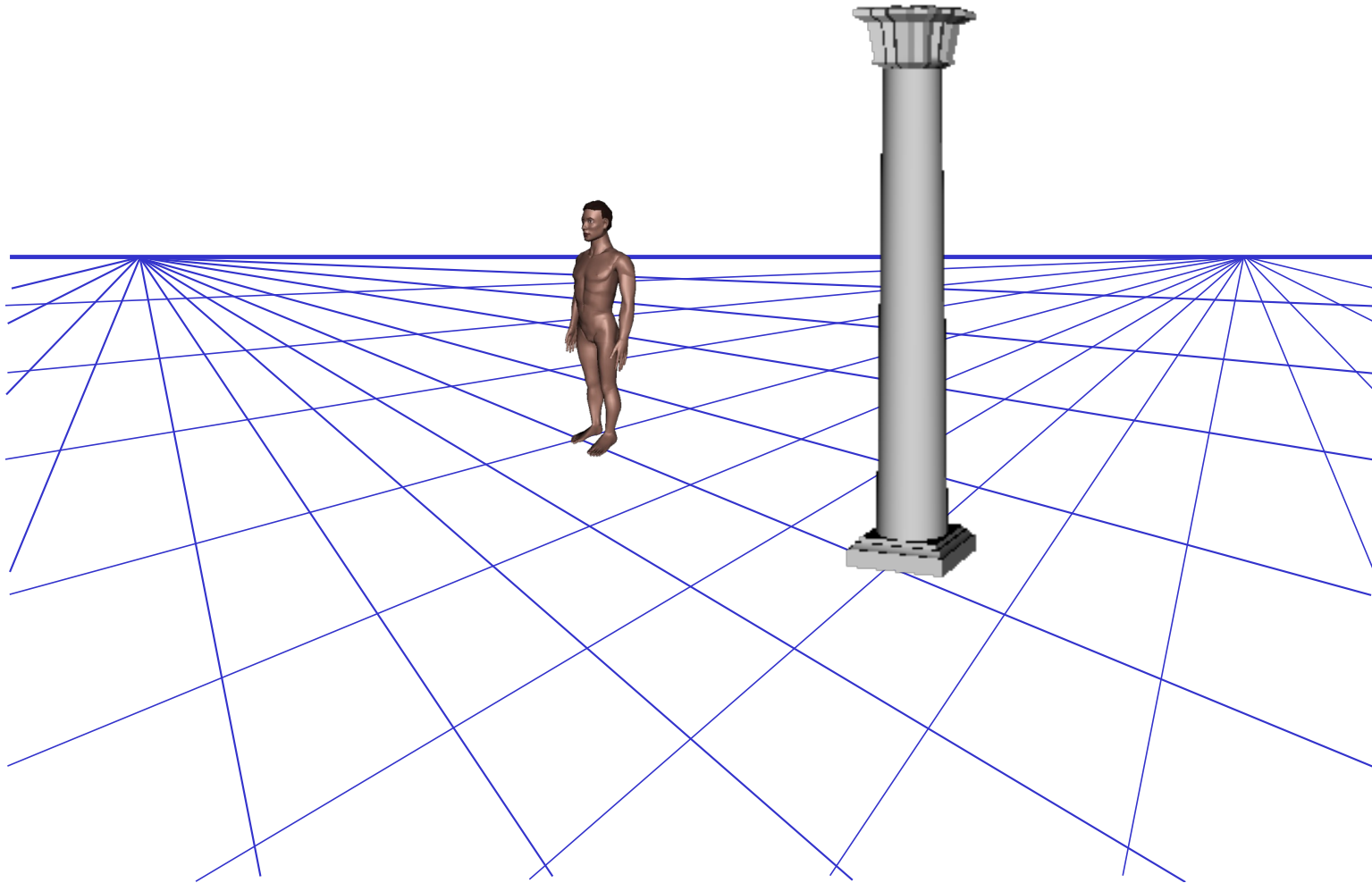
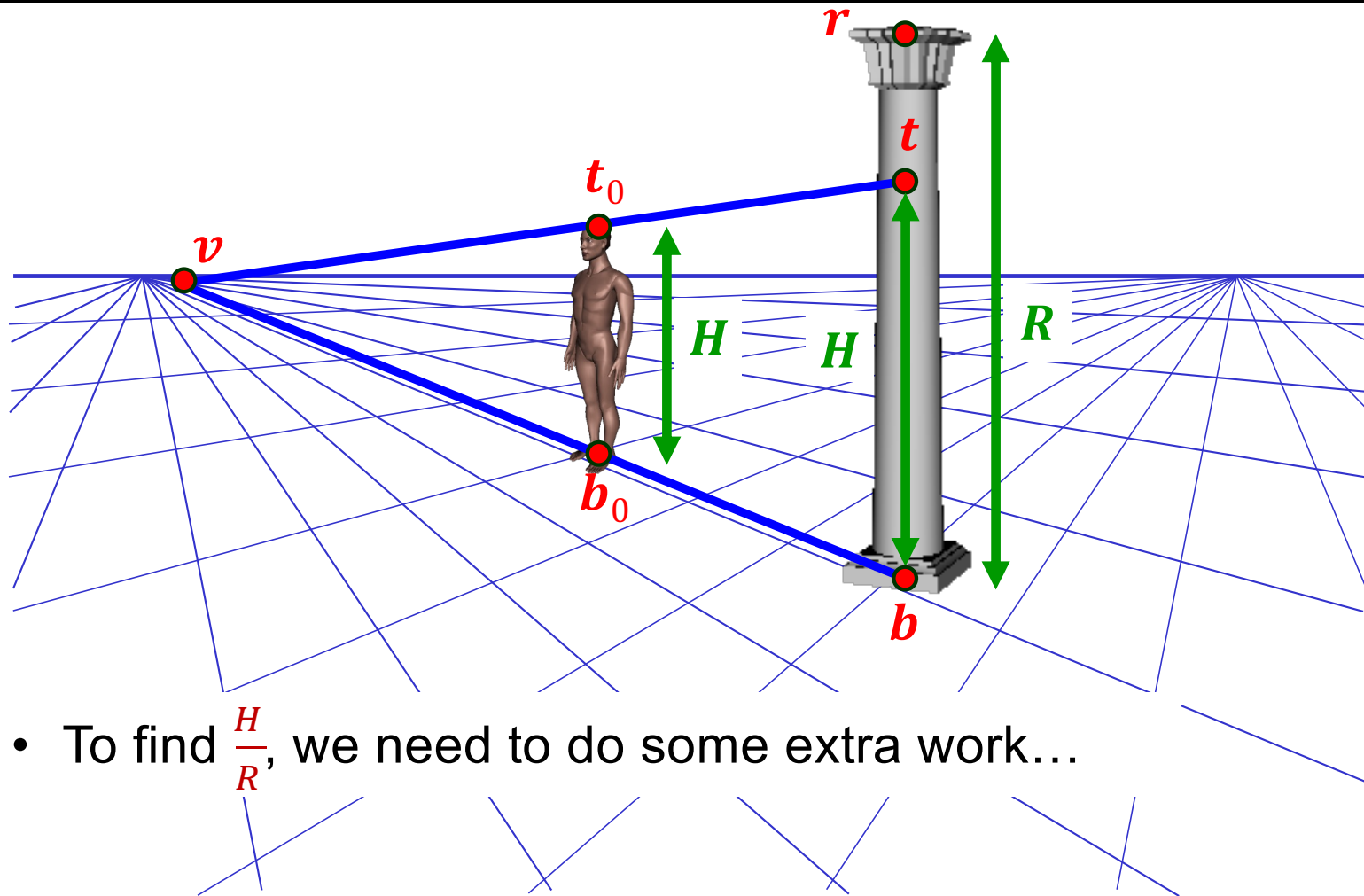


Image source: S. Seitz

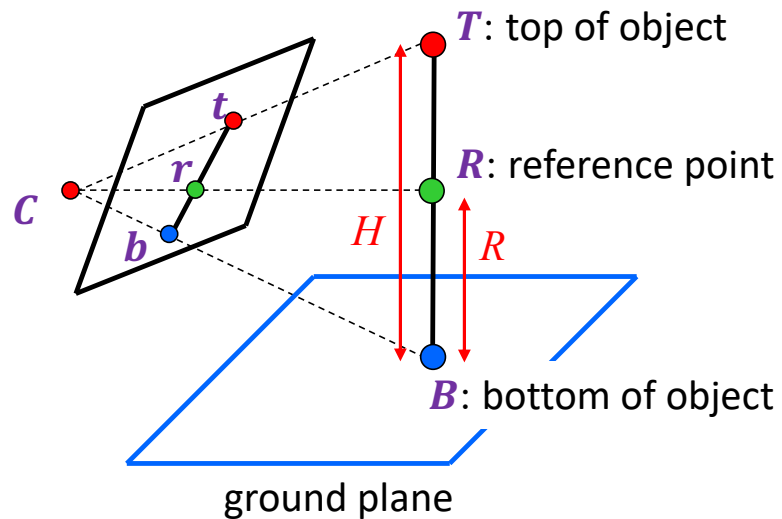
Measuring height without a ruler



Measuring height without a ruler



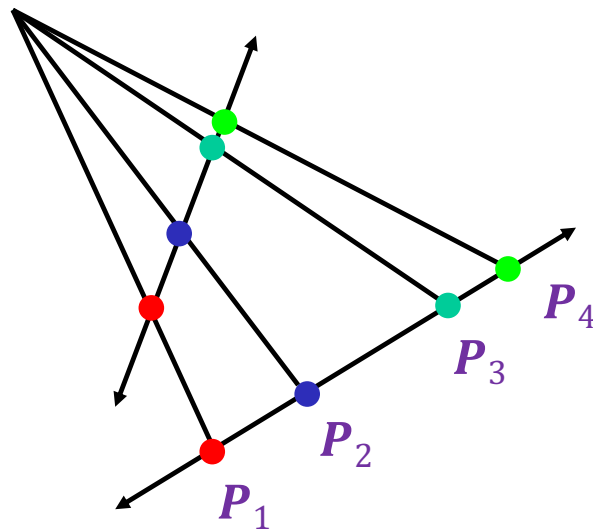
Measuring height



To make 3D measurements based on image observations, we need an **invariant** that does not change under perspective projection from 3D to 2D

Cross-ratio

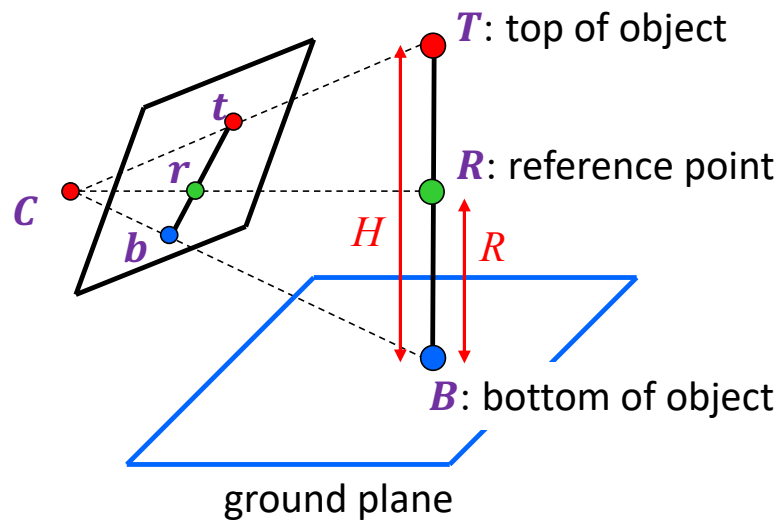
- Defined for four collinear points, invariant to projective transformations, including perspective projection from 3D to 2D
 - Also invariant to permutation of the points



$$\frac{\|P_3 - P_1\| \|P_4 - P_2\|}{\|P_3 - P_2\| \|P_4 - P_1\|}$$

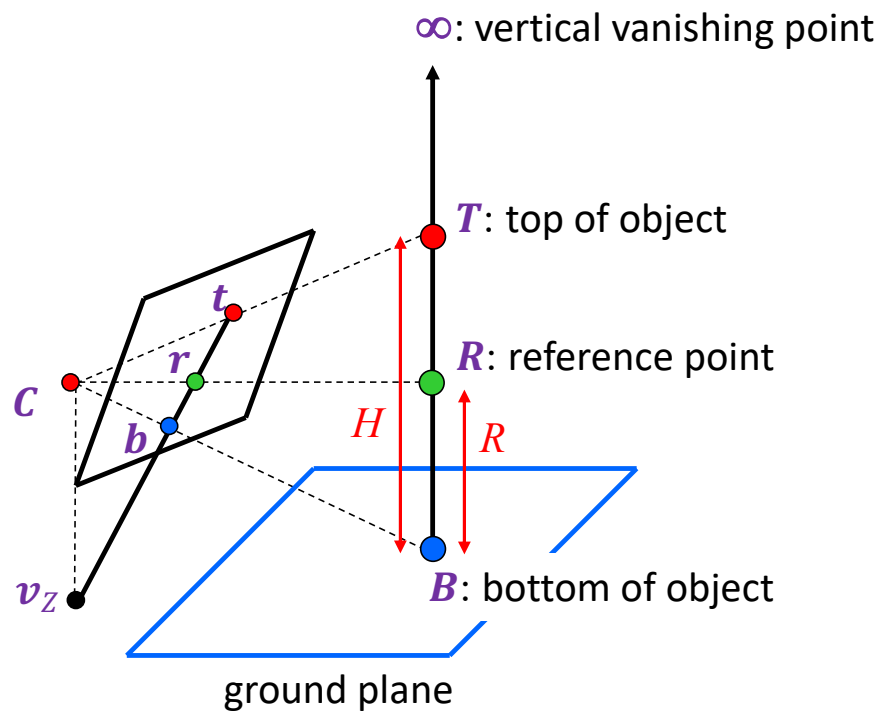
Measuring height

- We need a fourth point to compute a cross-ratio – what could it be?



Measuring height

- We need a fourth point to compute a cross-ratio – what could it be?



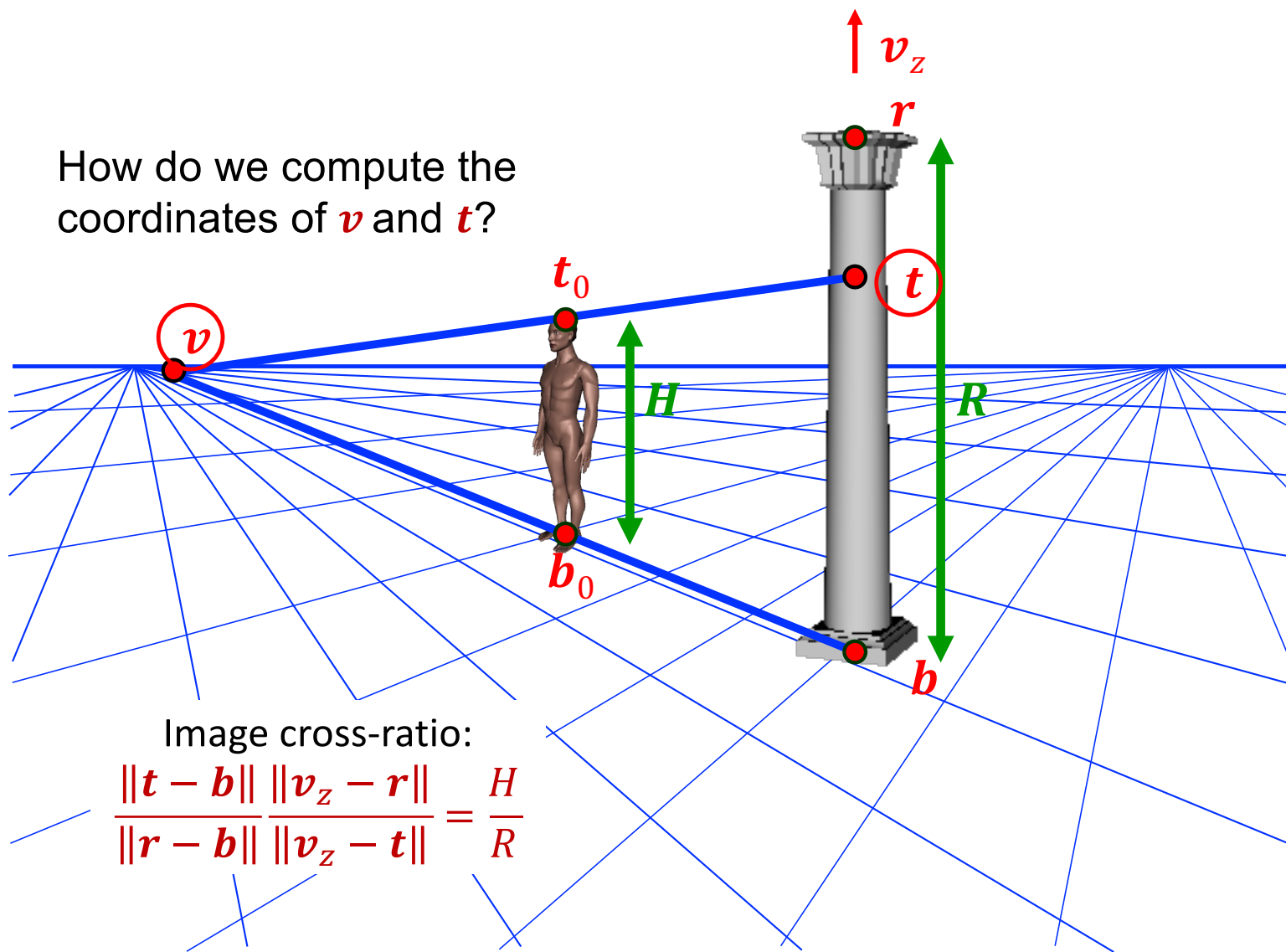
Scene cross ratio:

$$\frac{\|T - B\| \| \infty - R \|}{\|R - B\| \| \infty - T \|} = \frac{H}{R}$$

Image cross ratio:

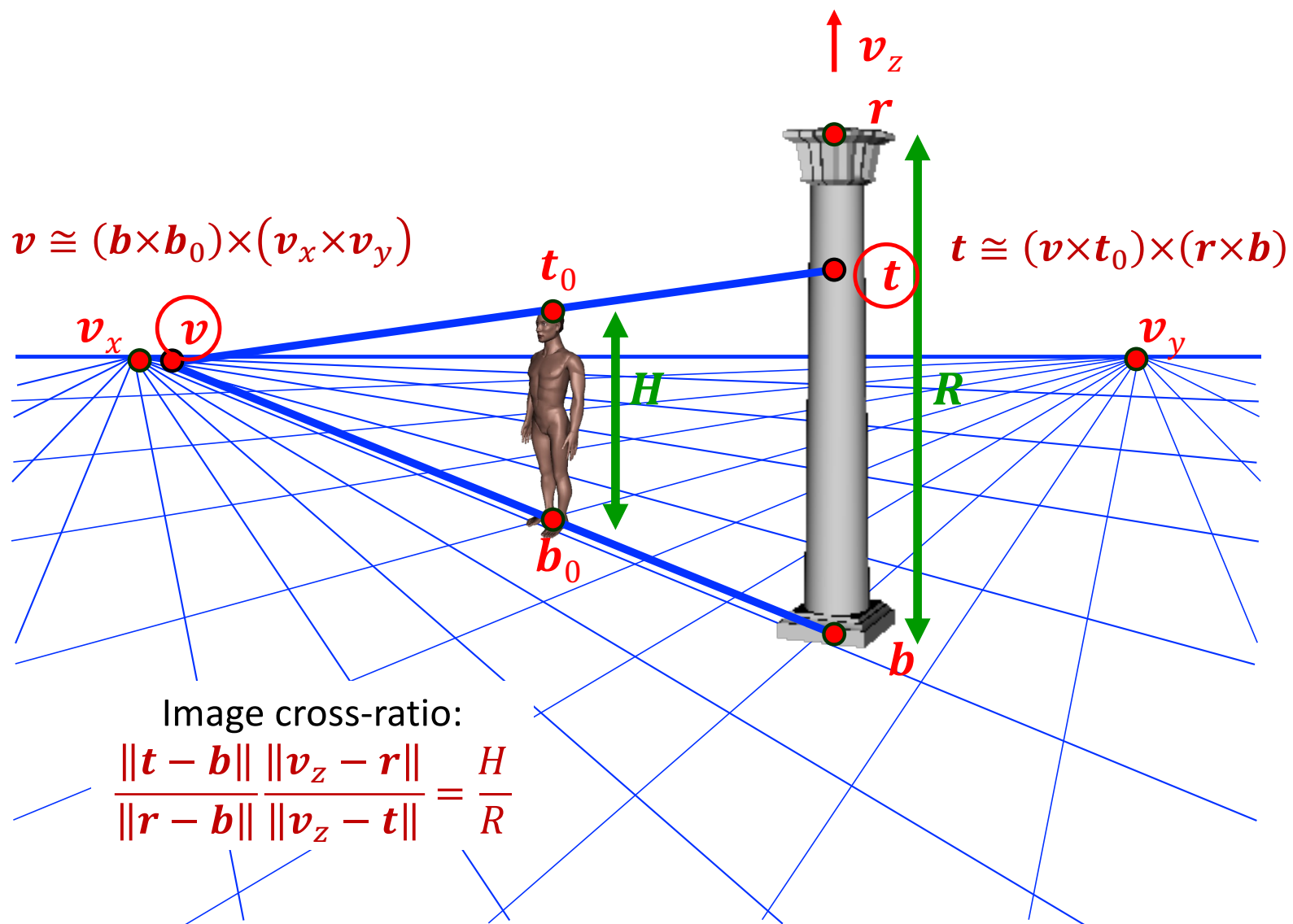
$$\frac{\|t - b\| \|v_z - r\|}{\|r - b\| \|v_z - t\|} = \frac{H}{R}$$

How do we compute the coordinates of v and t ?

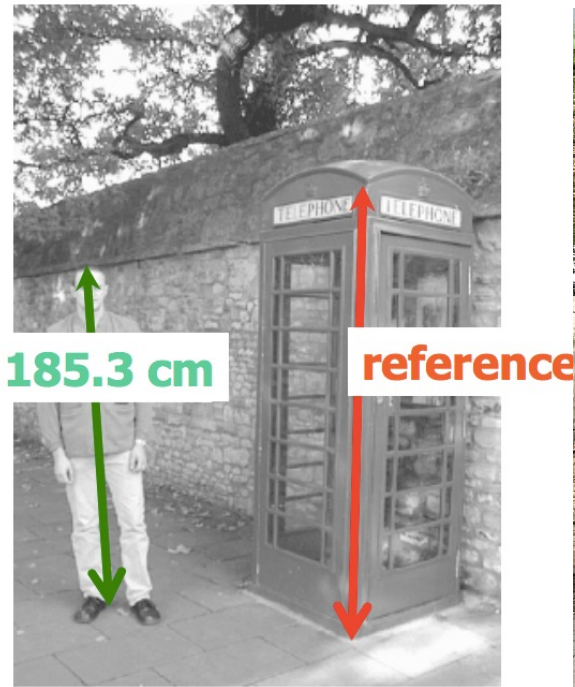


2D lines in homogeneous coordinates

- Line equation: $ax + by + c = 0$
- We can write this as $\mathbf{l}^T \mathbf{x} = 0$, where $\mathbf{l} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$
- How can we obtain the coefficient vector of line $\mathbf{l}$ passing through two points $\mathbf{x}_1$ and $\mathbf{x}_2$?
 - We want $\mathbf{l}$ s.t. $\mathbf{l}^T \mathbf{x}_1 = 0$, $\mathbf{l}^T \mathbf{x}_2 = 0$
 - This is given by $\mathbf{l} = \mathbf{x}_1 \times \mathbf{x}_2$
- What about point $\mathbf{x}$ that is the intersection of two lines $\mathbf{l}_1$ and $\mathbf{l}_2$?
 - $\mathbf{x} = \mathbf{l}_1 \times \mathbf{l}_2$
 - What happens if $\mathbf{l}_1$ and $\mathbf{l}_2$ are parallel?



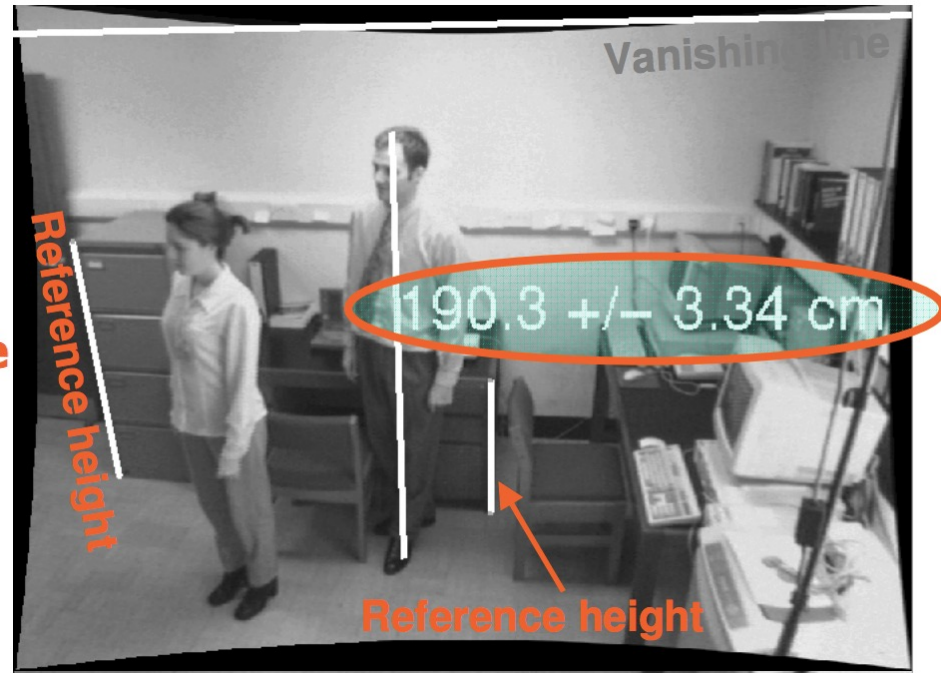
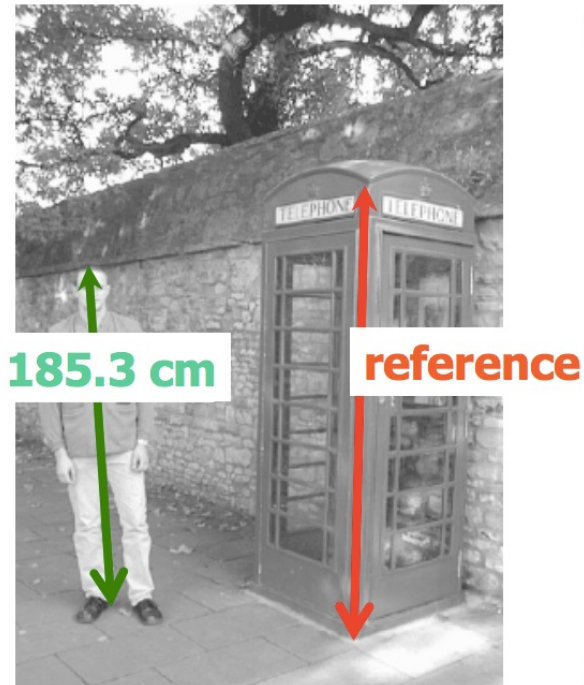
Height measurement examples



That booth is still there! (Oxford, September 2022)

A. Criminisi, I. Reid, and A. Zisserman, [Single View Metrology](#), IJCV 2000
Figure from [UPenn CIS580 slides](#)

Height measurement examples



A. Criminisi, I. Reid, and A. Zisserman, [Single View Metrology](#), IJCV 2000
Figure from [UPenn CIS580 slides](#)

Another example

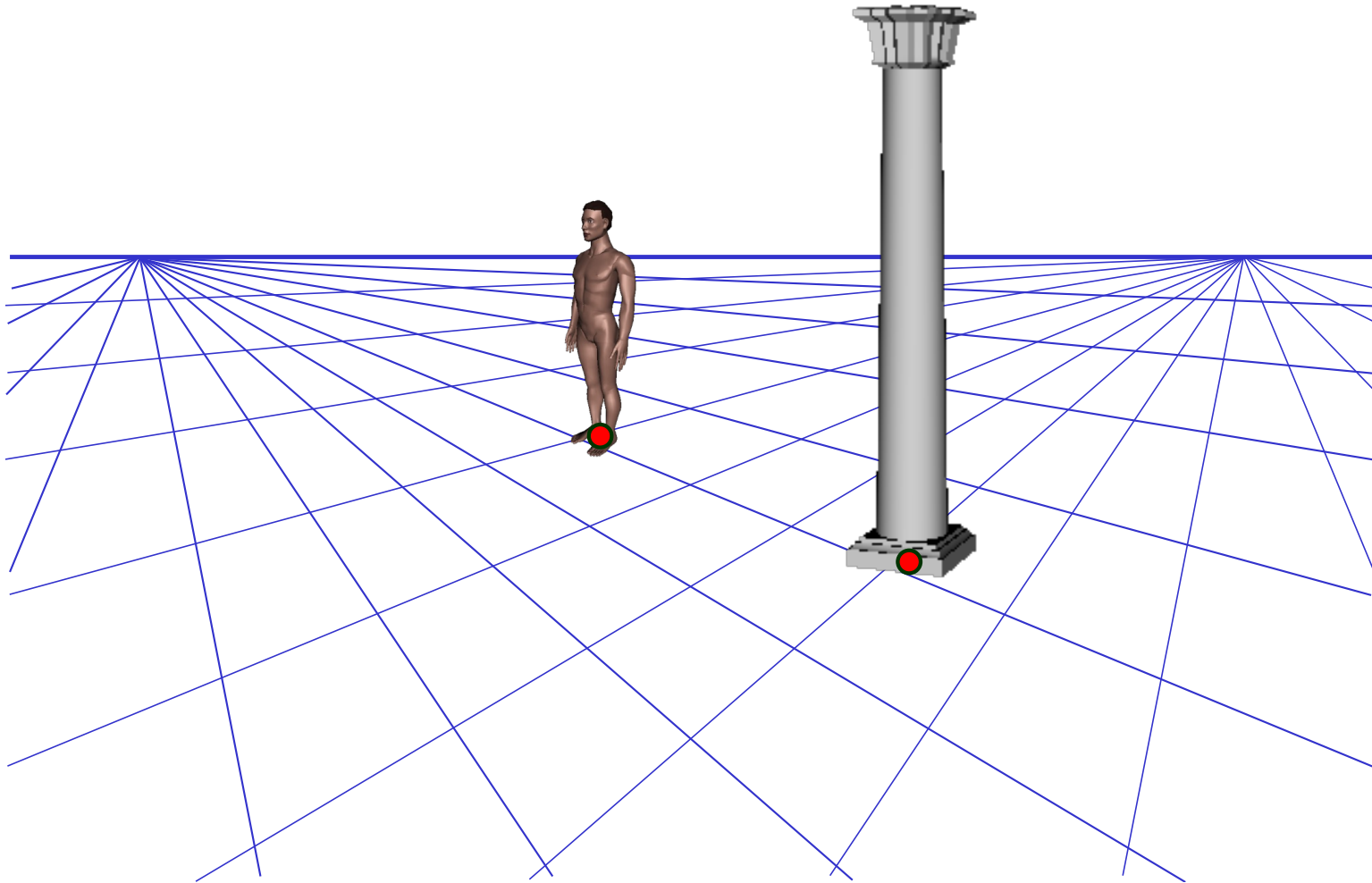
- Are the heights of the two groups of people consistent with one another?
- Measure heights using Christ as reference



Piero della Francesca, *Flagellation*, ca. 1455

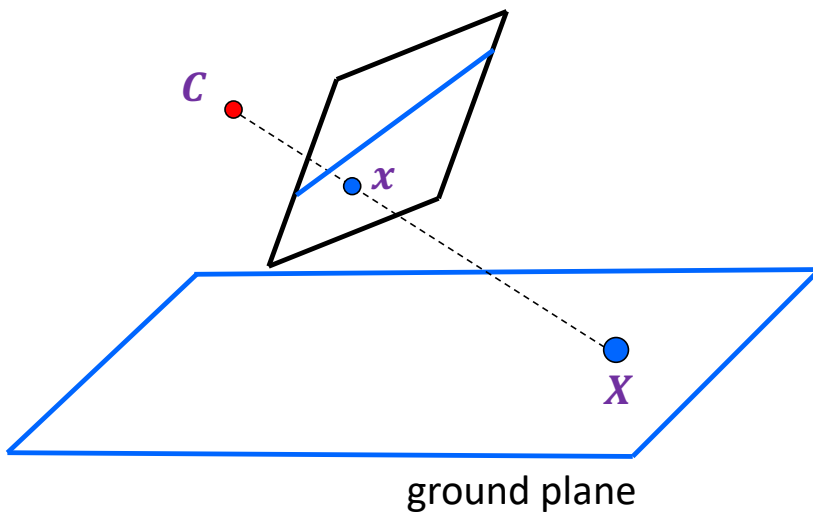
A. Criminisi, M. Kemp, and A. Zisserman, [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings.](#)
Proc. Computers and the History of Art, 2002

Locating a 3D point on the ground plane



Locating a 3D point on the ground plane

- Assuming that $\mathbf{x}$ is a projection of a point on the ground plane, how do we reconstruct $\mathbf{X}$?



- First, find the direction $\mathbf{x}_\infty$ in space corresponding to image point $\mathbf{x}$:

$$\mathbf{x} \cong \mathbf{K}[\mathbf{R}|\mathbf{0}] \begin{pmatrix} \mathbf{x}_\infty \\ 0 \end{pmatrix}$$

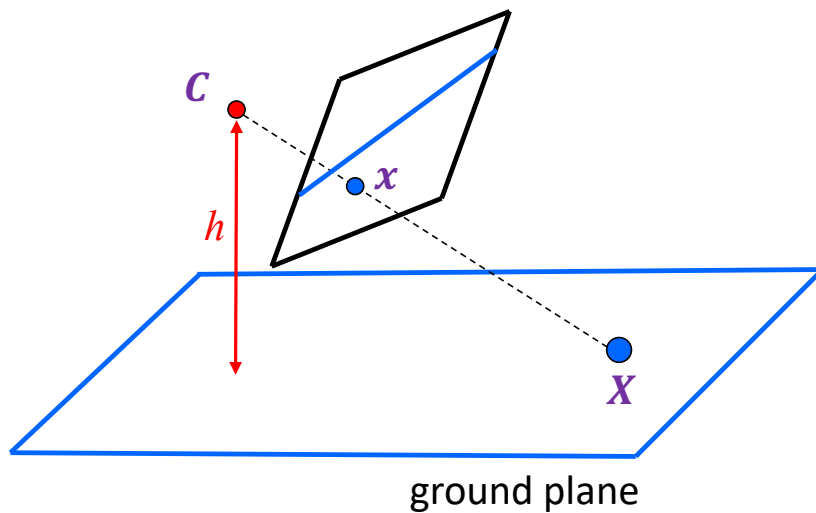
$$\mathbf{x} \cong \mathbf{K}\mathbf{R}\mathbf{x}_\infty$$

$$\mathbf{x}_\infty \cong \mathbf{R}^T \mathbf{K}^{-1} \mathbf{x}$$

- Recall: knowing the camera calibration allows us to map points in the image to visual ray directions in space!

Locating a 3D point on the ground plane

- Assuming that x is a projection of a point on the ground plane, how do we reconstruct X ?

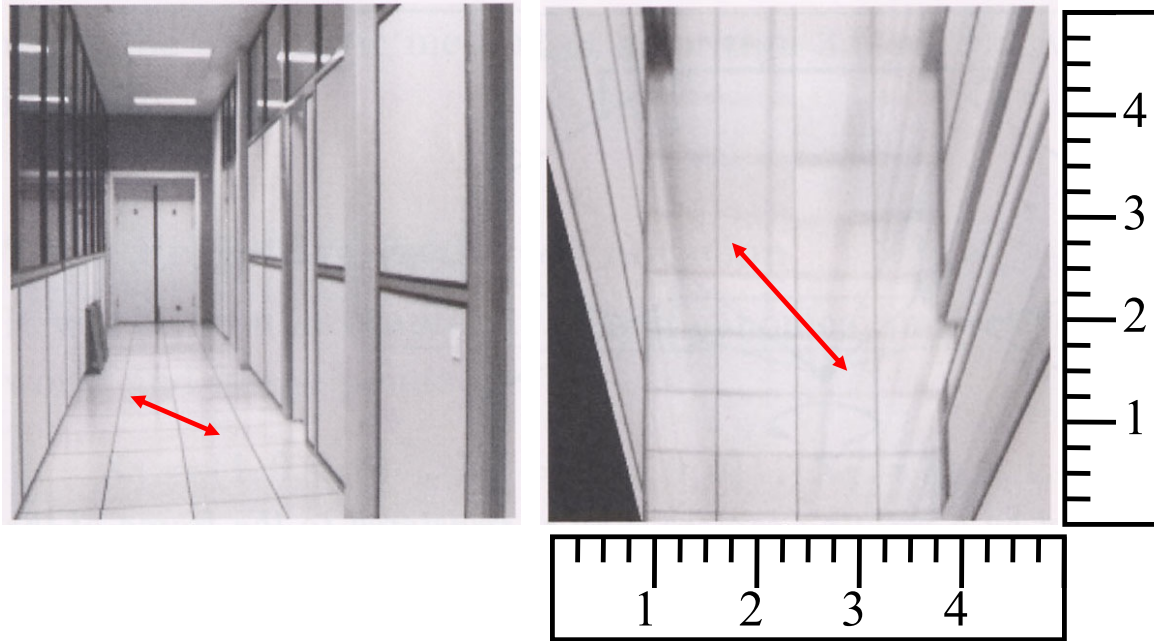


- First, find the direction x_{∞} in space corresponding to image point x :

$$x_{\infty} \cong R^T K^{-1} x$$

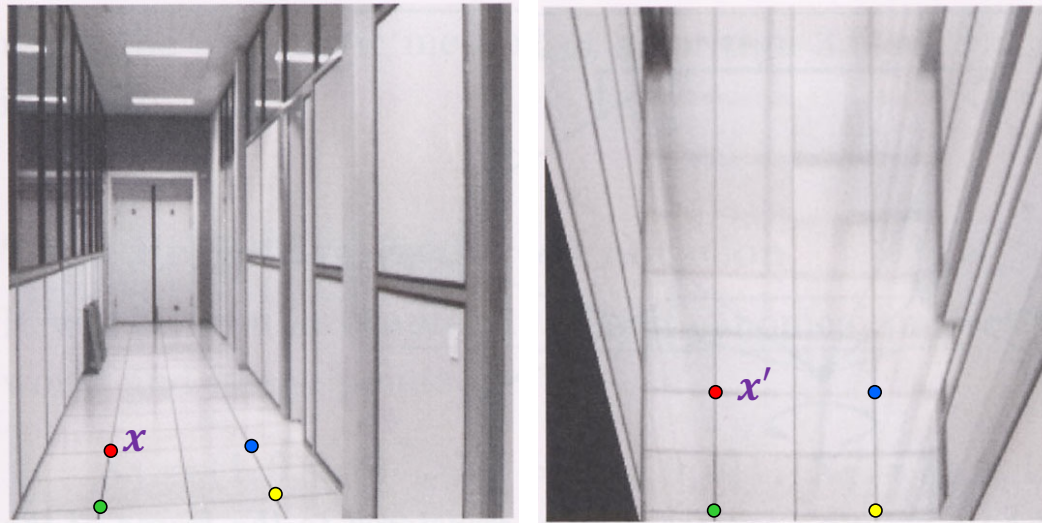
- The vertical vanishing direction is the normal to the plane. Let's say it's the z direction. Then the equation of the ground plane is $z = -h$, where h is the height of the camera above the ground plane
- Finally, all we need to do is scale x_{∞} so that its third entry is equal to $-h$, and append a 1 to get X

Measurements within planes



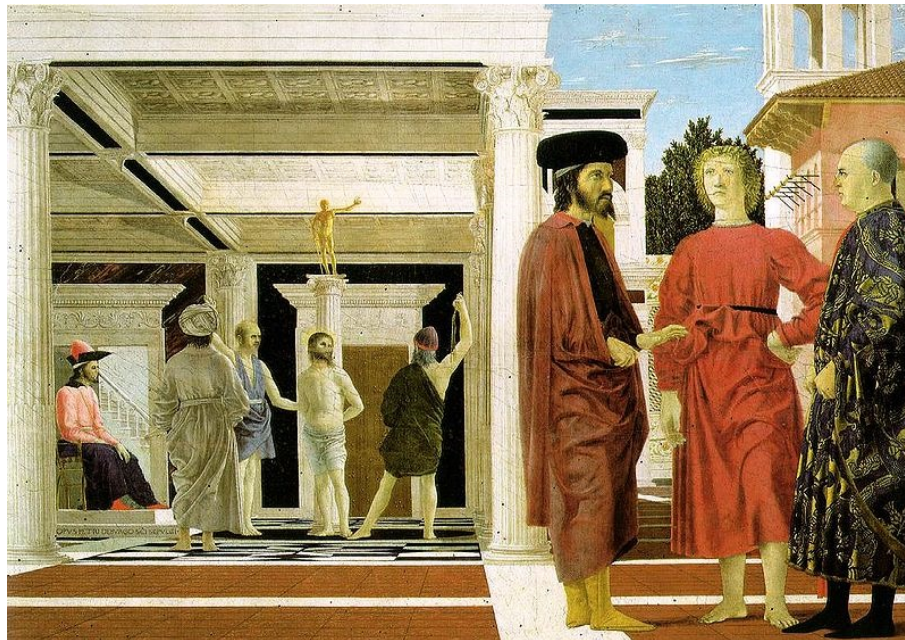
- Simplest approach: unwarp then measure

Image rectification

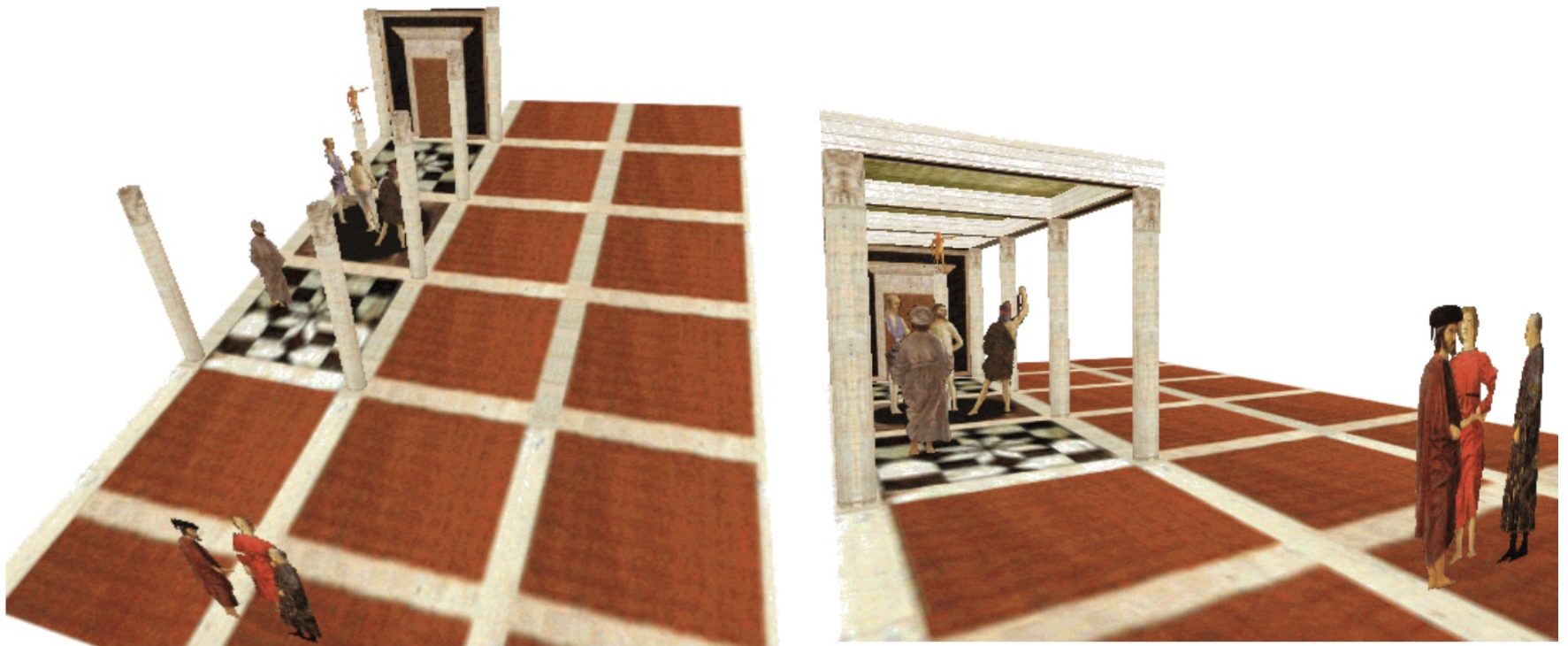


- To unwarp (rectify) an image, solve for homography H given four pairs of matches assumed to have a known configuration (e.g., square)

Image rectification: Example



Putting everything together: Single-view modeling

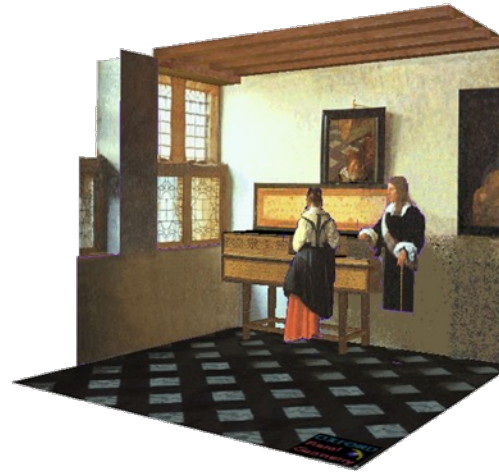


A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings.](#)
Proc. Computers and the History of Art, 2002

Putting everything together: Single-view modeling

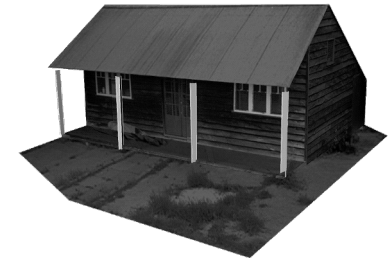
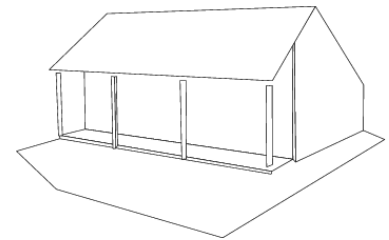
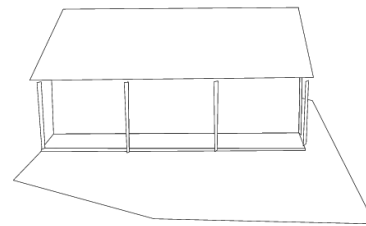


J. Vermeer, Music Lesson, 1662



A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings.](#)
Proc. Computers and the History of Art, 2002

Putting everything together: Single-view modeling

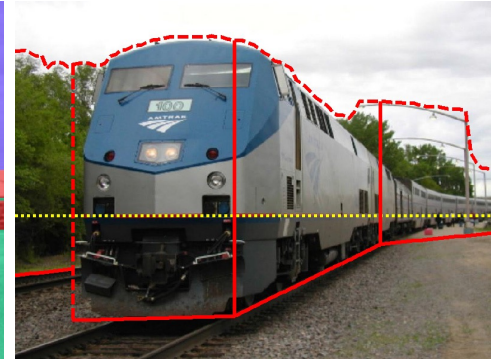
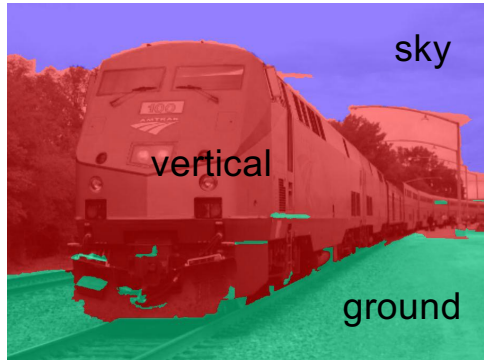


A. Criminisi et al. [Single View Metrology](#). IJCV 2000

Outline

- Vanishing points and lines
- Camera calibration using vanishing points
- Measurements from a single image
- Applications of single-view metrology

Application: Fully automatic modeling



Automatic Photo Pop-up

D. Hoiem A.A. Efros M. Hebert
Carnegie Mellon University

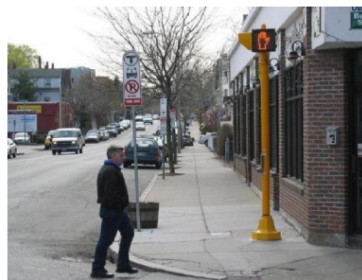
D. Hoiem, A.A. Efros, and M. Hebert, [Automatic Photo Pop-up](#), SIGGRAPH 2005

Application: Object detection

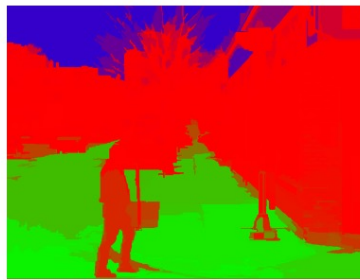


D. Hoiem, A.A. Efros, and M. Hebert. [Putting Objects in Perspective](#). CVPR 2006

Application: Object detection



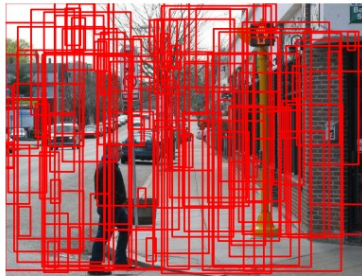
(a) input image



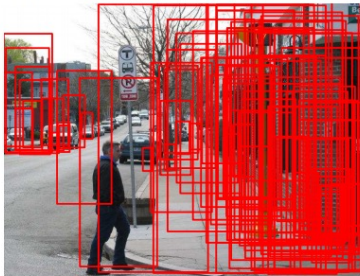
(c) surface orientation estimate



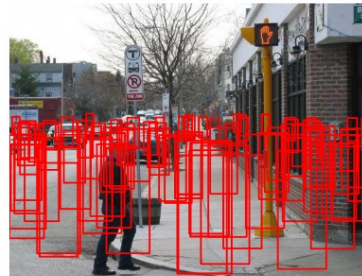
(e) $P(\text{viewpoint} \mid \text{objects})$



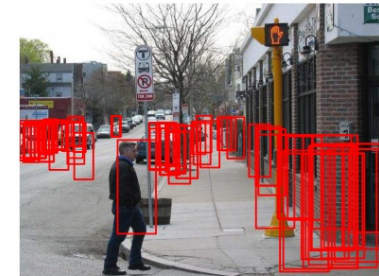
(b) $P(\text{person}) = \text{uniform}$



(d) $P(\text{person} \mid \text{geometry})$



(f) $P(\text{person} \mid \text{viewpoint})$



(g) $P(\text{person} \mid \text{viewpoint}, \text{geometry})$

D. Hoiem, A.A. Efros, and M. Hebert. [Putting Objects in Perspective](#). CVPR 2006

Application: Image editing

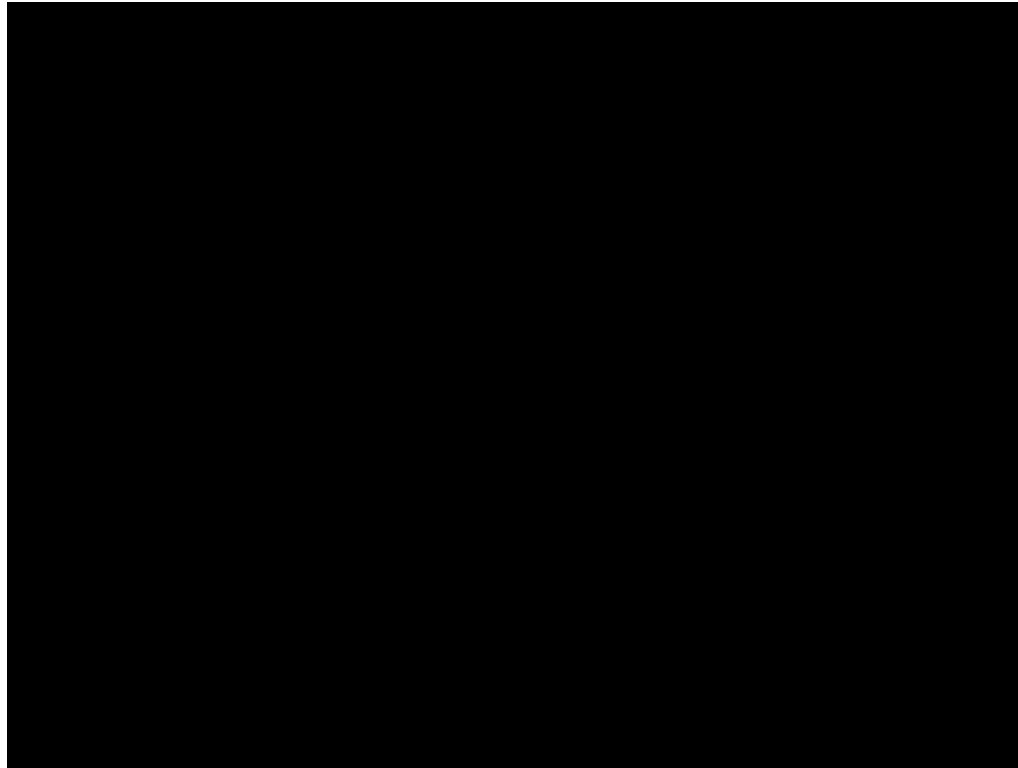
- Inserting synthetic objects into images



K. Karsch and V. Hedau and D. Forsyth and D. Hoiem. [Rendering Synthetic Objects into Legacy Photographs](#). *SIGGRAPH Asia* 2011

Application: Image editing

- Inserting synthetic objects into images:



K. Karsch and V. Hedau and D. Forsyth and D. Hoiem. [Rendering Synthetic Objects into Legacy Photographs](#). *SIGGRAPH Asia* 2011